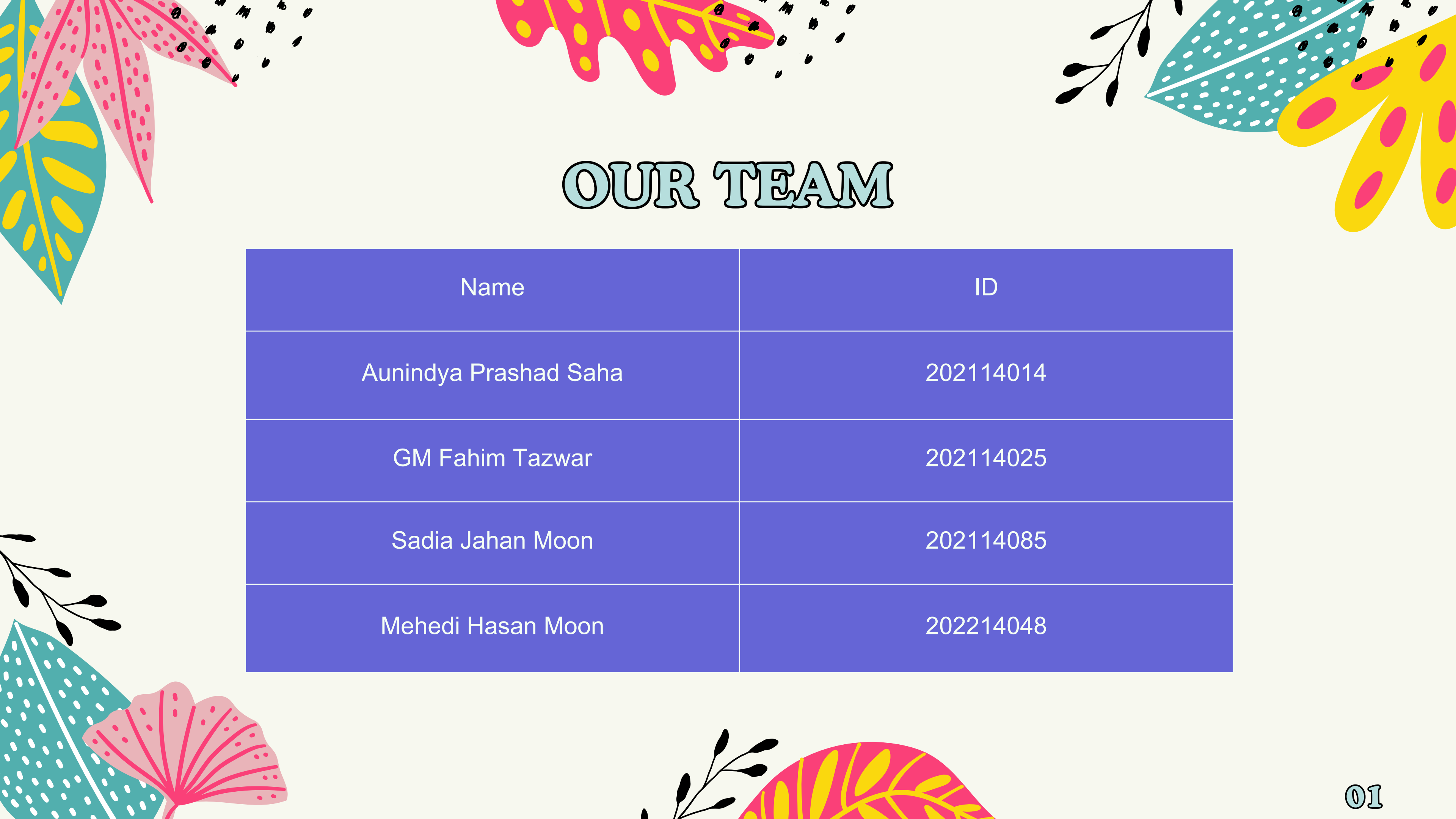


CAMPUSNET

An Interactive University Community
Platform for Students & Smart
Assistance

By Group 4





OUR TEAM

Name	ID
Aunindya Prashad Saha	202114014
GM Fahim Tazwar	202114025
Sadia Jahan Moon	202114085
Mehedi Hasan Moon	202214048

INTRODUCTION

CampusNet is an interactive mobile application built using Flutter. The app connects students, alumni, and faculty members, creating a unified platform for campus life. It provides essential services such as an AI-powered chatbot, campus navigation map, messaging system, and emergency features such as blood donation ambulance services etc. The app aims to help new students connect with their peers and alumni and help them navigate the campus more effectively.





TARGET USERS:

The app is designed for a wide range of users including:

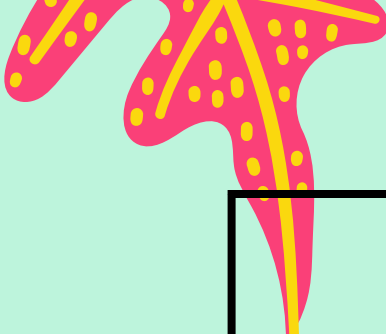
- Undergraduate Students
- Graduate Students
- Alumni



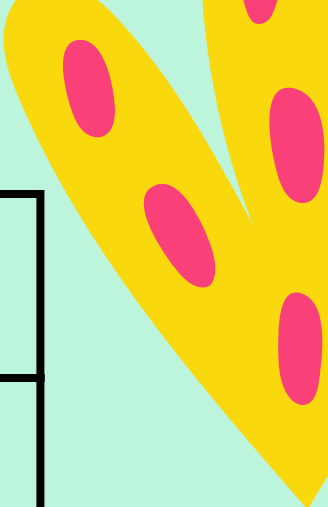
PROBLEM STATEMENT AND IDENTIFICATION:

Based on the review of existing systems, the primary issues identified are:

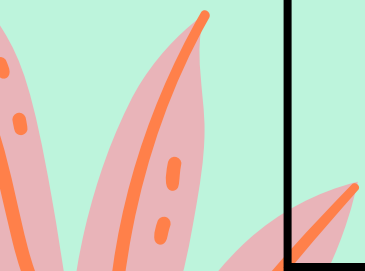
- Lack of a unified platform for campus-related services.
 - Inadequate connection mechanisms for students and alumni.
 - Fragmentation of essential services like campus navigation, emergency services, and tuition opportunities across different apps.
 - Lack of AI-driven support for campus-related questions.
- 



LITERATURE REVIEW



App	Features	Limitations
Campus.pk	Video lectures, MCQ tests, live sessions	Limited to Pakistan; lacks campus-specific services
Campus+ (FH Campus Wien)	Class schedules, campus card balance, cafeteria menu	Limited to FH Campus Wien students
Campus Student	Real-time access to grades, assignments, attendance	Requires institutional lo- gin; U.S.-centric
Campus.uno	ERP solution for academic and ad- ministrative pro- cesses	Primarily an administra- tive tool; lacks student en- gagement features





OBJECTIVES

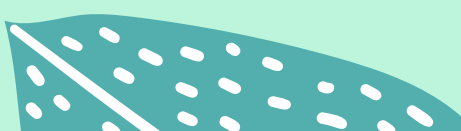
The main objectives of the CampusNet app are:

- To provide instant assistance and campus guidance through an integrated AI chatbot.
 - To build a unified platform for communication, emergency services, and alumni-student networking.
 - To enhance campus navigation and accessibility with an interactive map.
- 



ETHICAL CONCERNS

The app raises several ethical concerns, including:

- User data (personal, academic) is collected and stored securely with user consent.
 - Location data is used exclusively for campus navigation with explicit user approval.
 - Blood donation and emergency service features require user consent to share contact details.
 - No microphone or camera access is requested unless the user voluntarily uses the voice input feature for the chatbot.
- 



FEATURE LIST



1. Chatbot: AI-powered assistant for answering questions, giving campus info, and guiding students anytime.
 2. Campus Map: Interactive map to navigate buildings, classrooms, events, and facilities on campus.
 3. Blog post: A platform for students and clubs to post updates, events, and stories with easy browsing.
 4. Resource Bank : Provides all study materials required for all courses of all departments
 5. Tuition: provides tuition opportunity to students
 6. Message: Single-chat ,Group chat. Group has a Resource page
 7. Emergency : contact with ambulance, blood donor, donate blood, request for blood .
- 
- 



TECH STACK:

Component	Tools Used
Frontend Framework	Dart
Backend Framework	Flask
Database	Postges
Additional Tools	Groq API

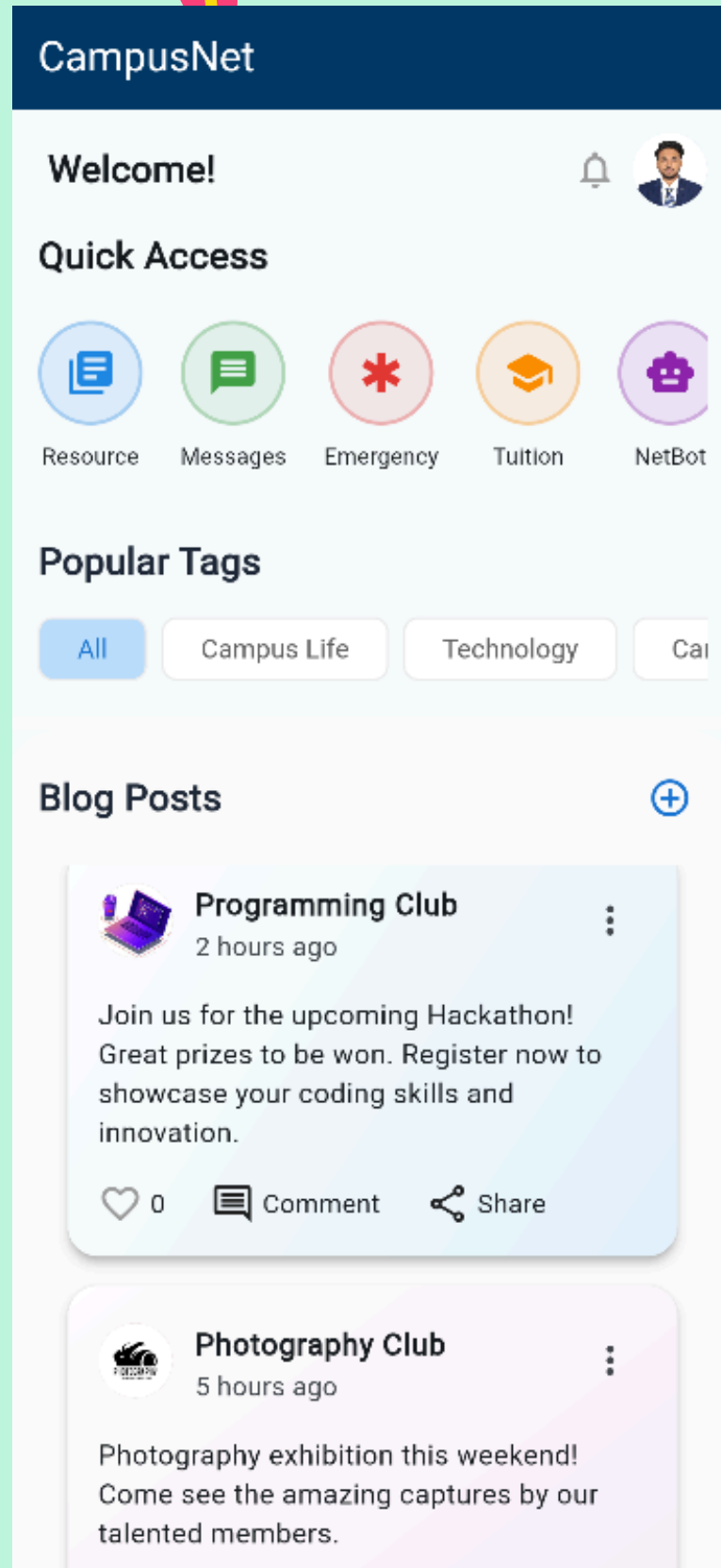


SOCIETAL IMPACTS:

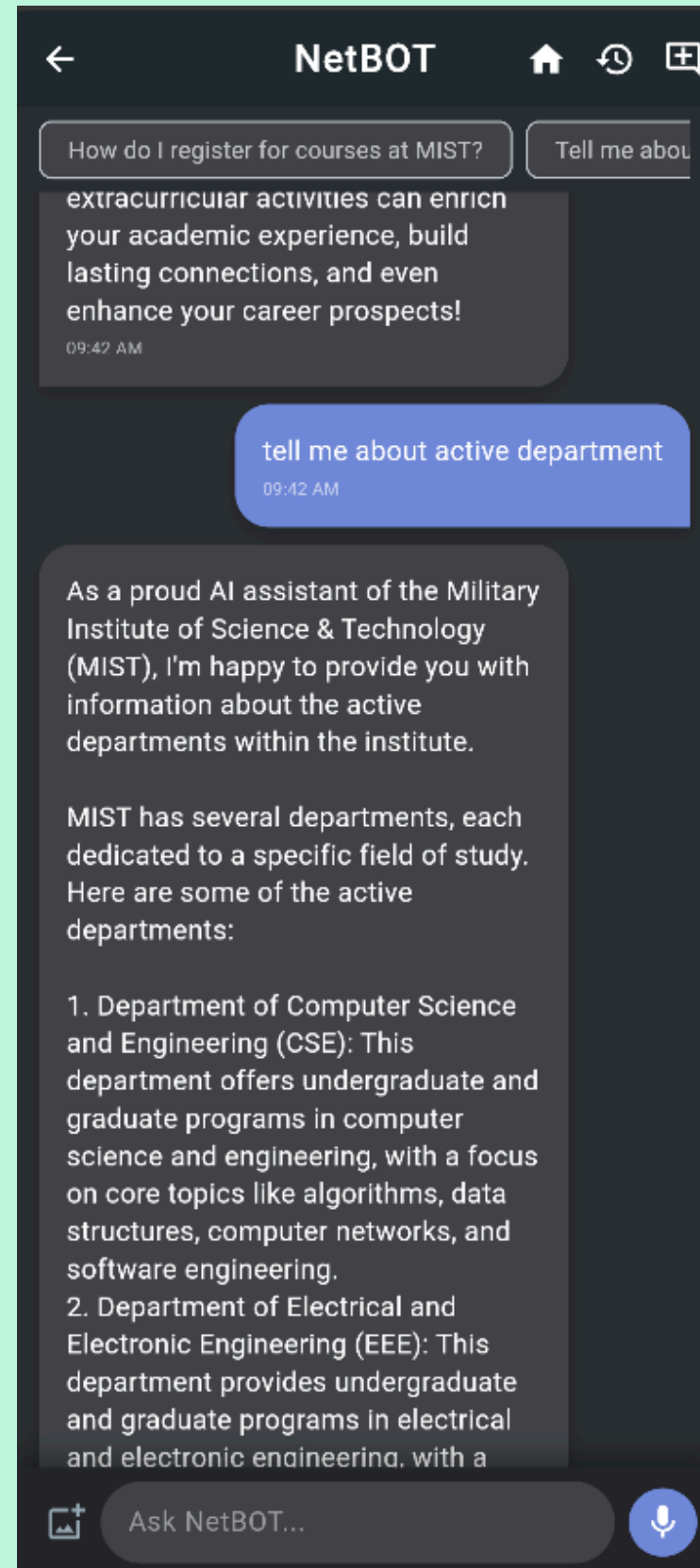
CampusNet can have a significant positive impact:

- Helps freshers navigate campus and connect with peers quickly.
 - Fosters a sense of community by enabling alumni-student mentorship.
 - Promotes social responsibility through features like blood donation and tutoring opportunities.
 - Enables students to access and share study materials/resources effortlessly.
- 
- 

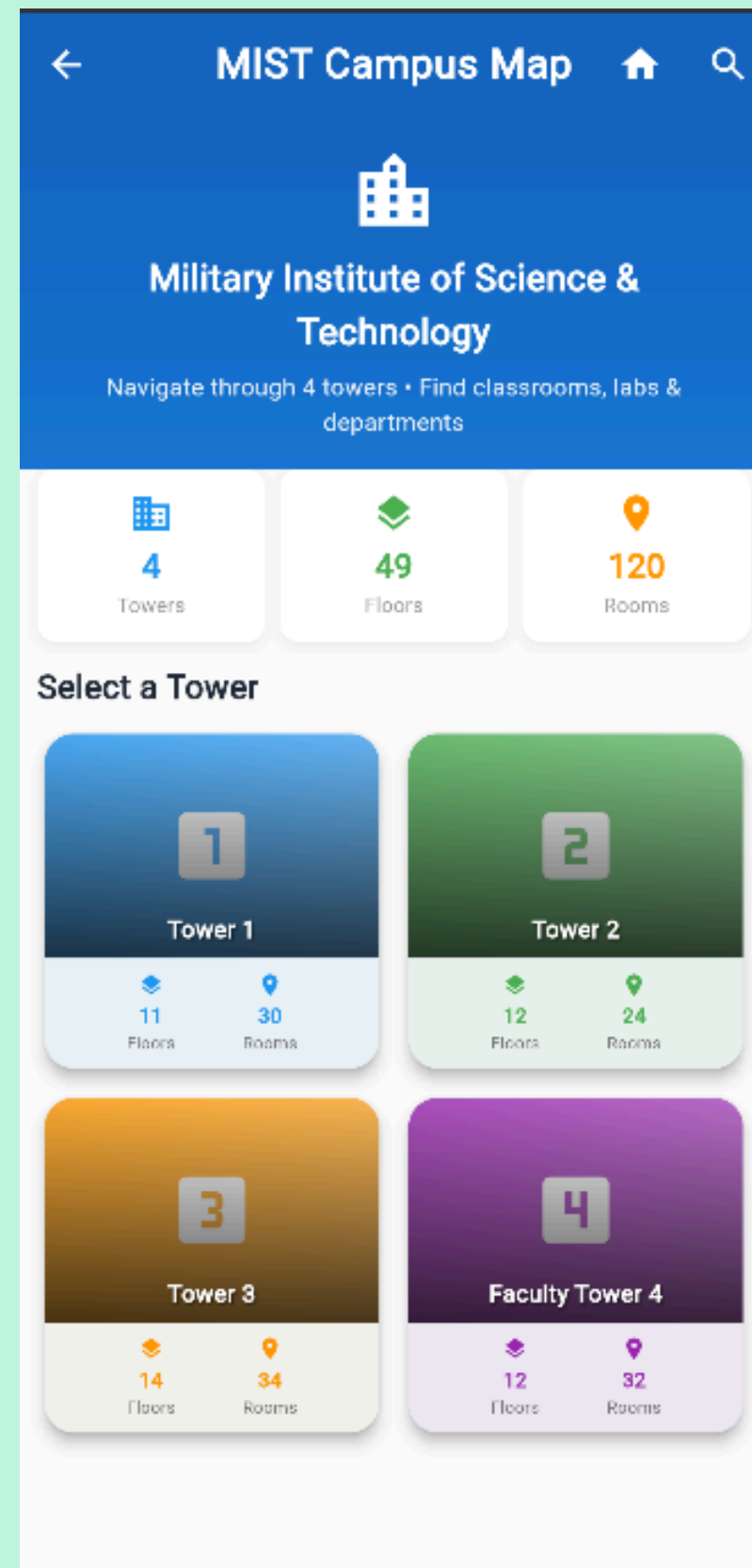
DEVELOPED SYSTEM



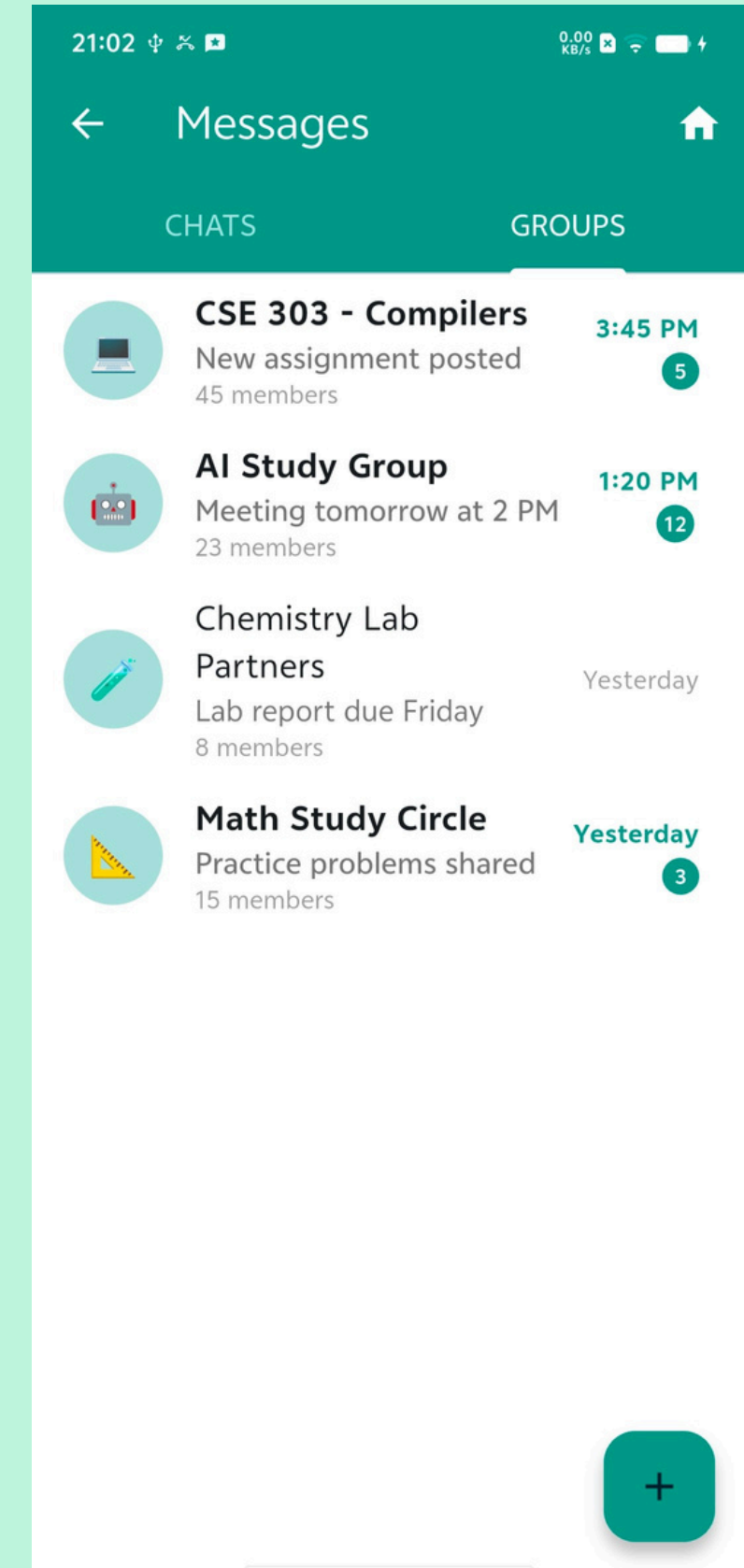
Home Page



ChatBot page

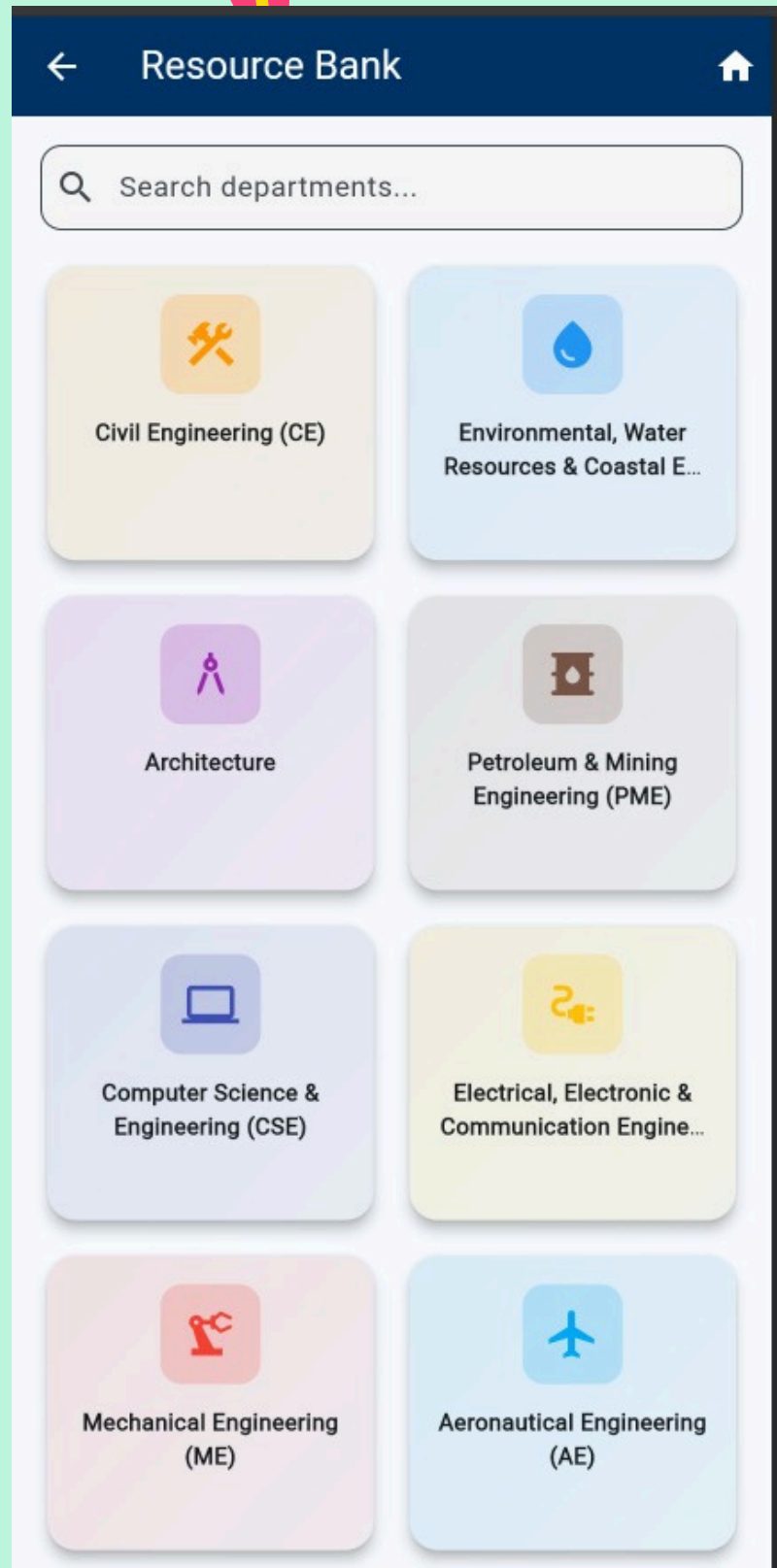


Campus Map

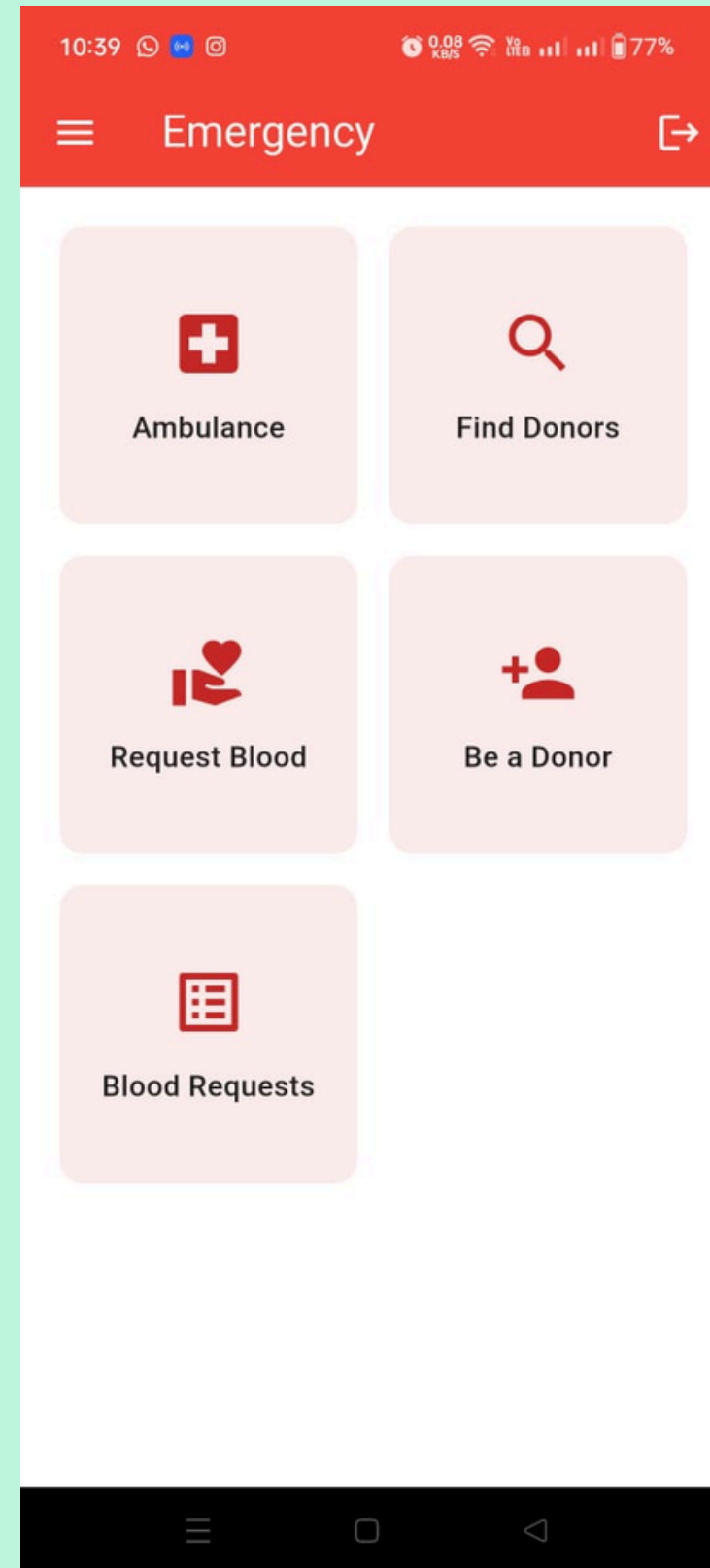


Message

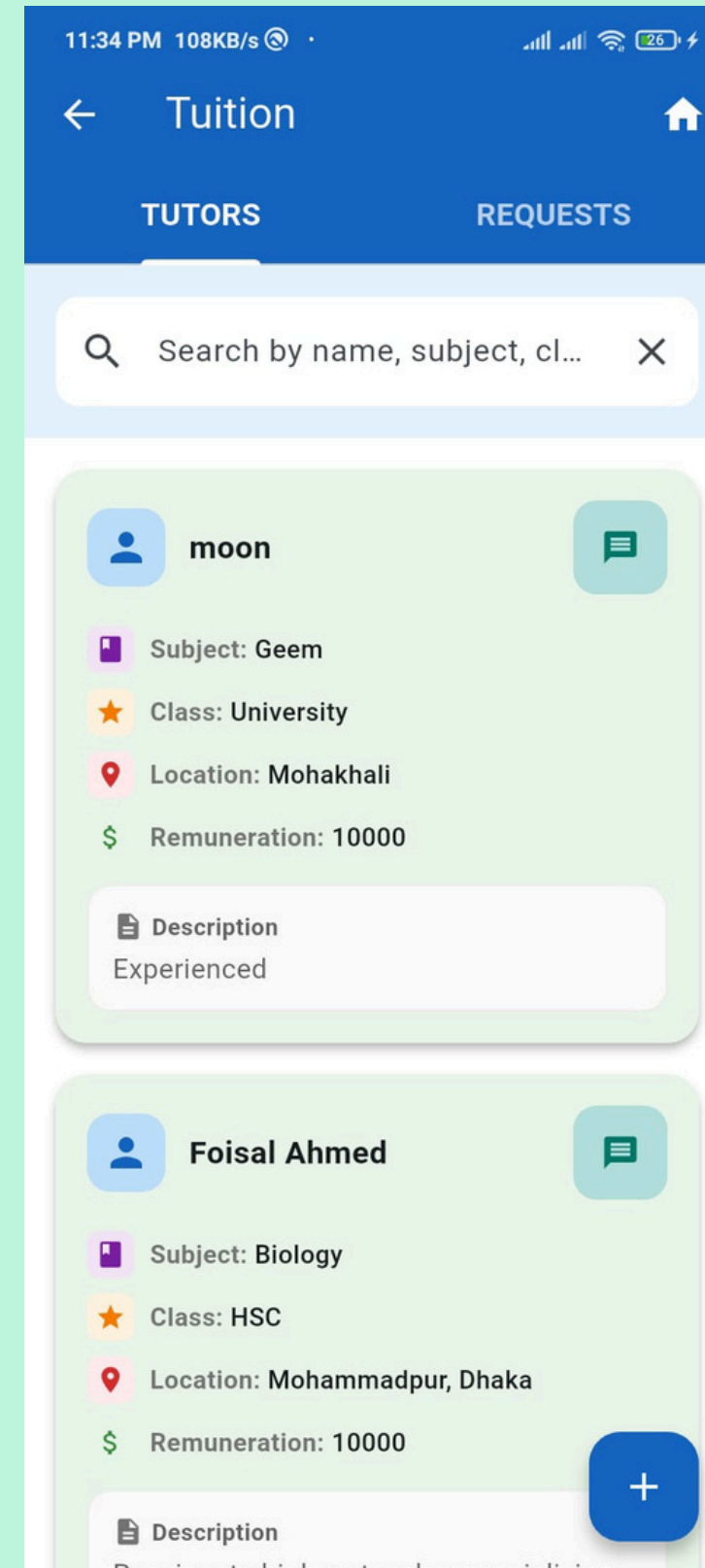
DEVELOPED SYSTEM



Resource Bank



Emergency Page



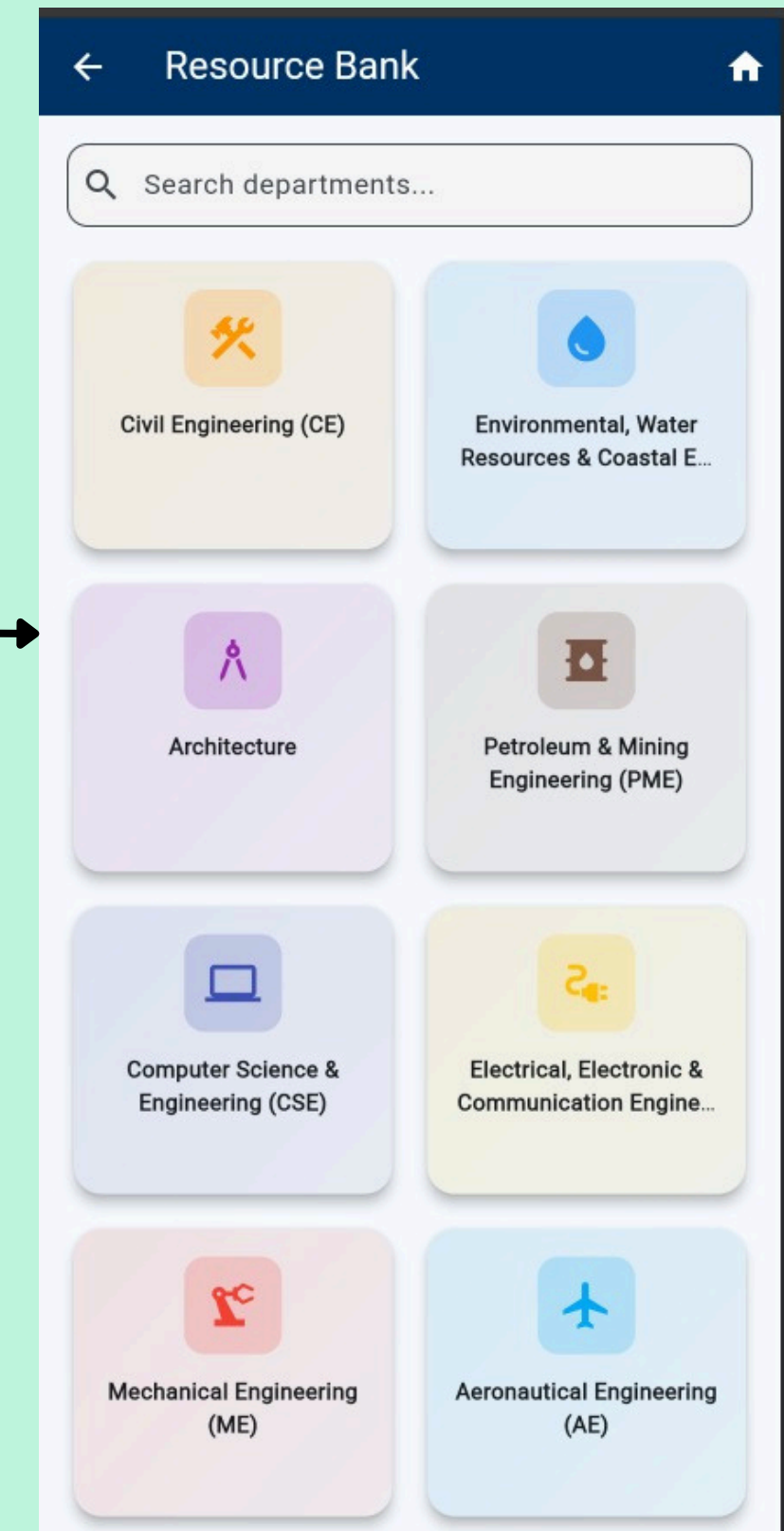
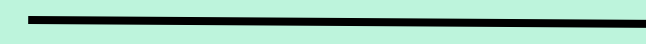
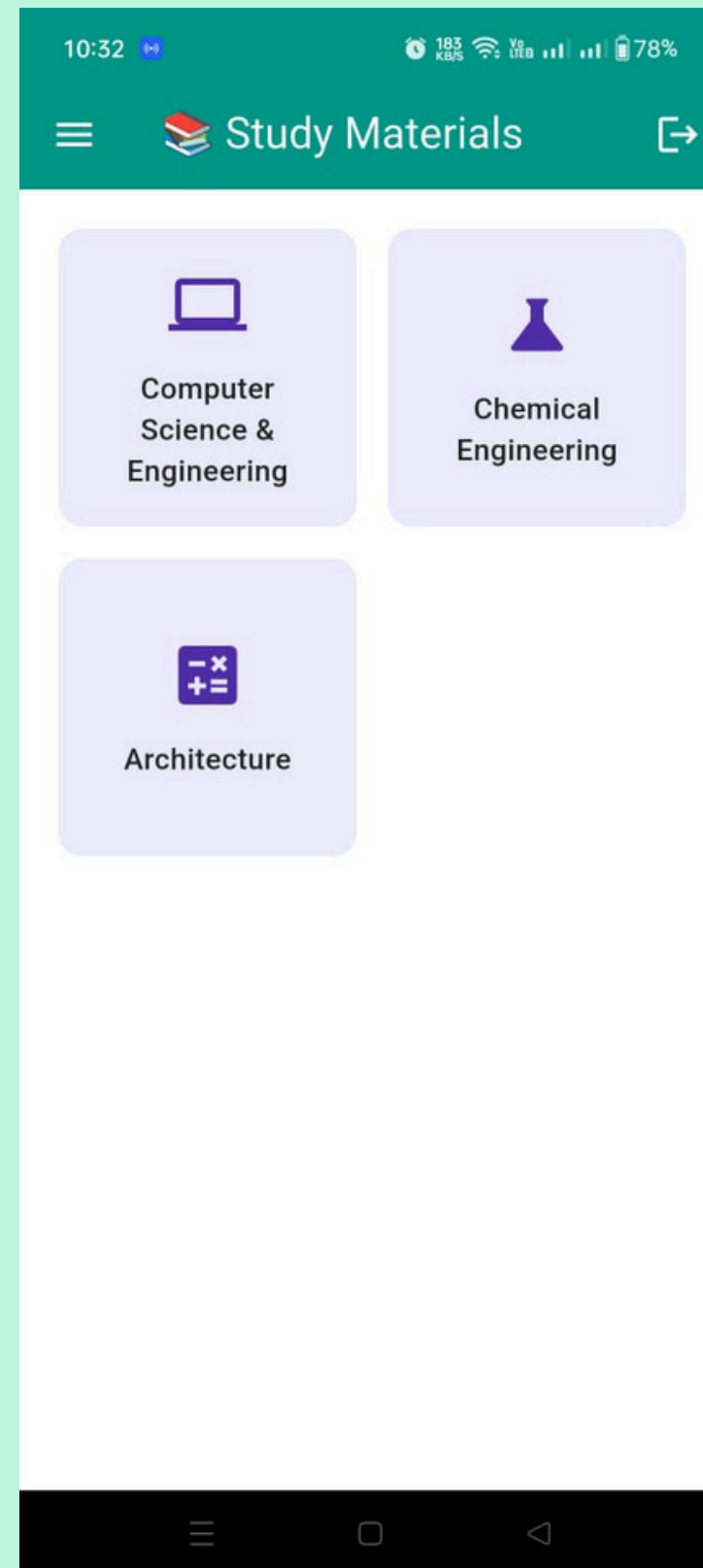
Tuition Page

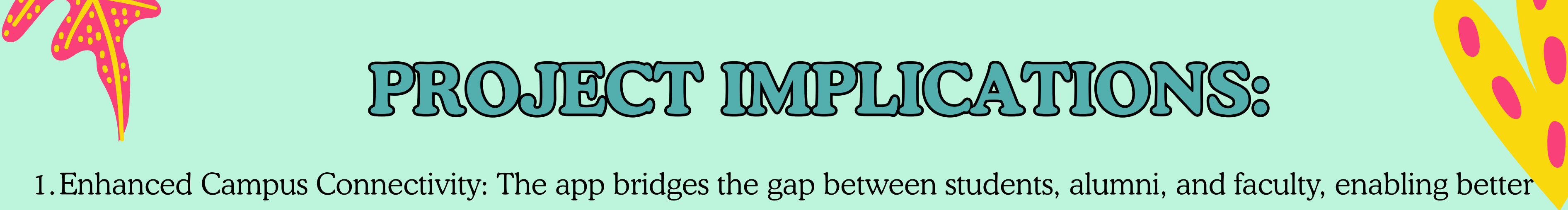


Profile

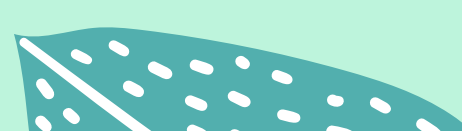
DESIGN CHANGES

Improved UI for the Resource Bank by adding a search option and enhancing the color scheme.



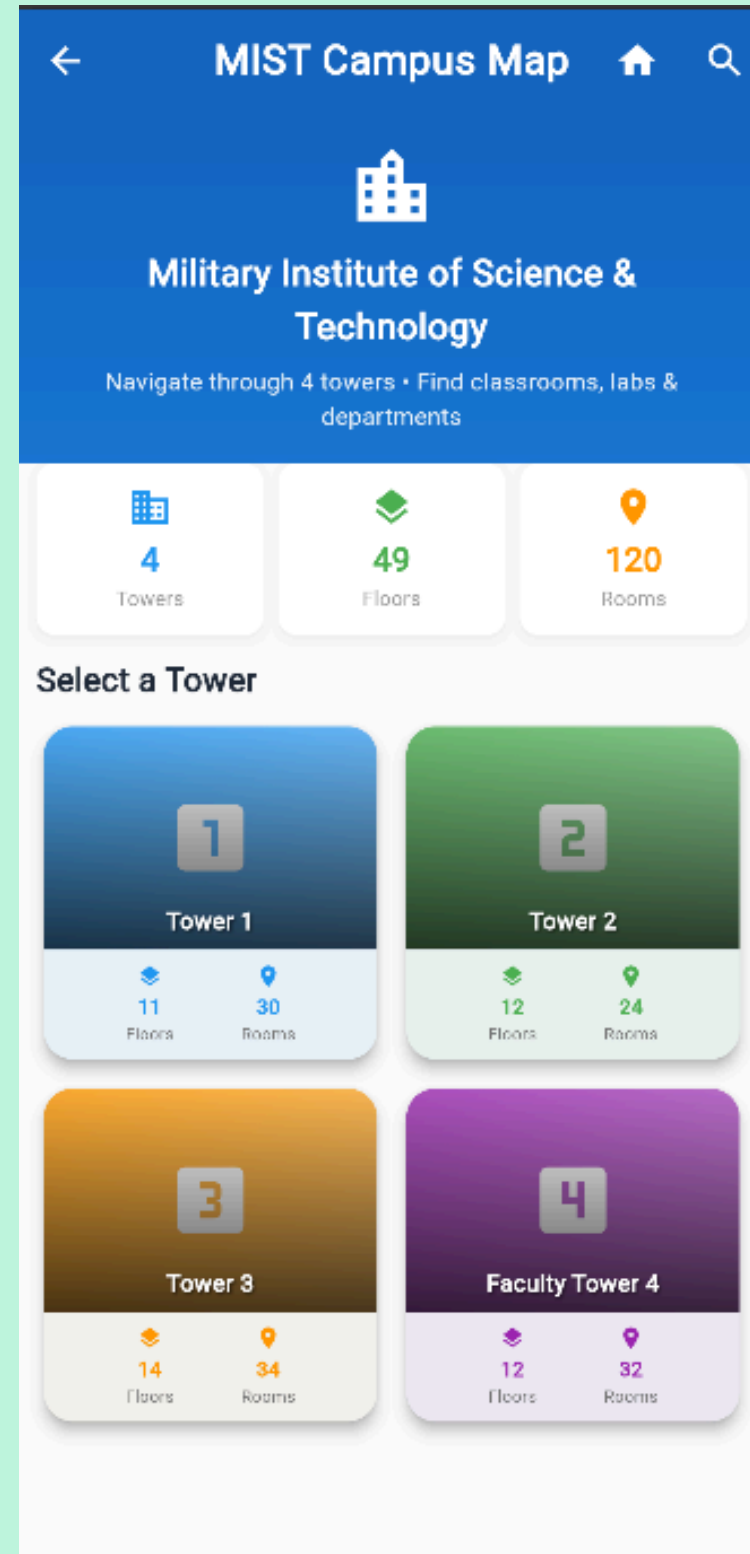


PROJECT IMPLICATIONS:

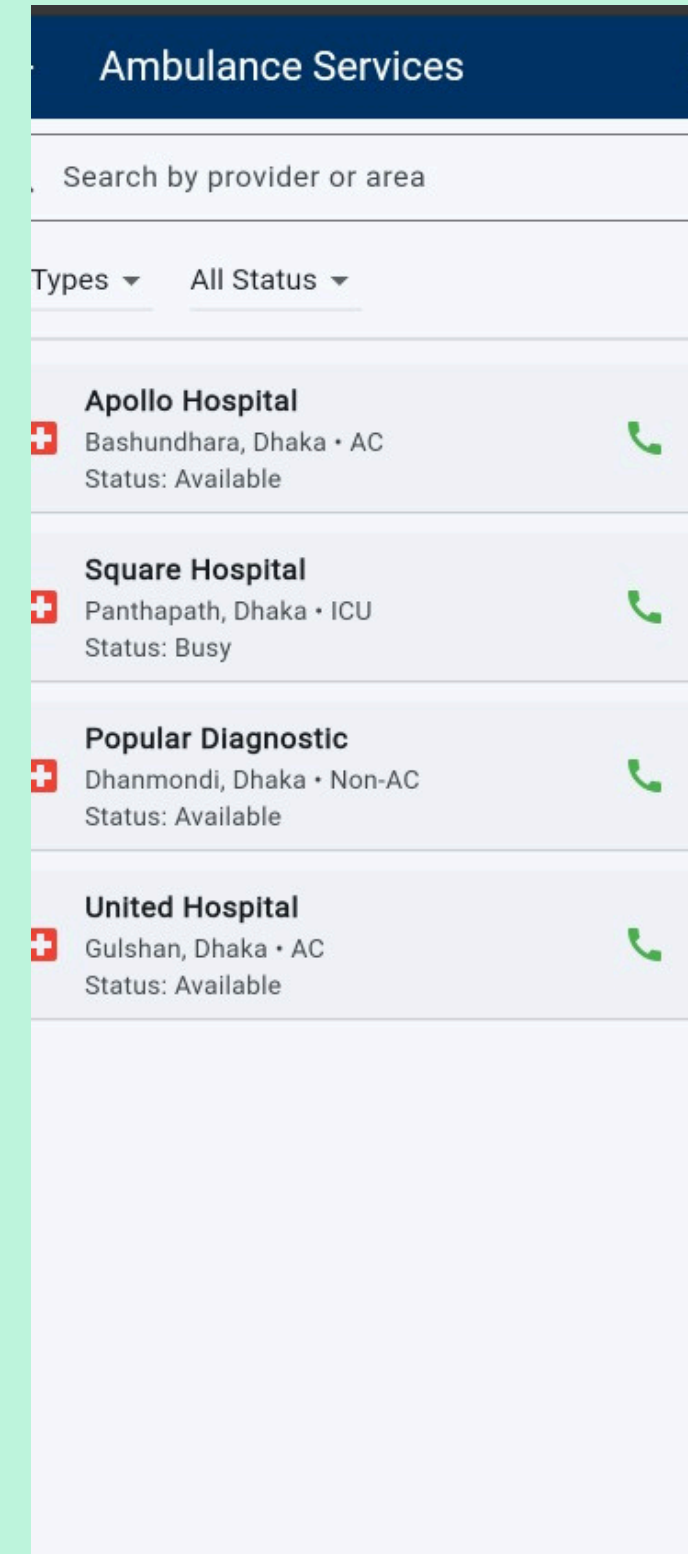
1. Enhanced Campus Connectivity: The app bridges the gap between students, alumni, and faculty, enabling better communication and access to resources.
 2. Increased Student Engagement: With features like blog posts, profiles, and tuition services, the app fosters active involvement in the campus community.
 3. Improved Campus Navigation: The interactive campus map provides easy access to essential locations, helping new students adapt quickly to campus life.
 4. Efficient Resource Sharing: The Resource Bank feature ensures that students have access to academic materials and resources, improving learning outcomes.
 5. Alumni Networking: Alumni can stay connected with current students, offering mentorship and career opportunities, strengthening the university community.
 6. Emergency Services Access: The integration of emergency services such as blood donation and ambulance contacts improves safety and response time on campus.
 7. Scalable Platform: The app's structure allows for the addition of new features and services, ensuring that it can grow with the needs of the university community.
- 

DESIGN CHANGES

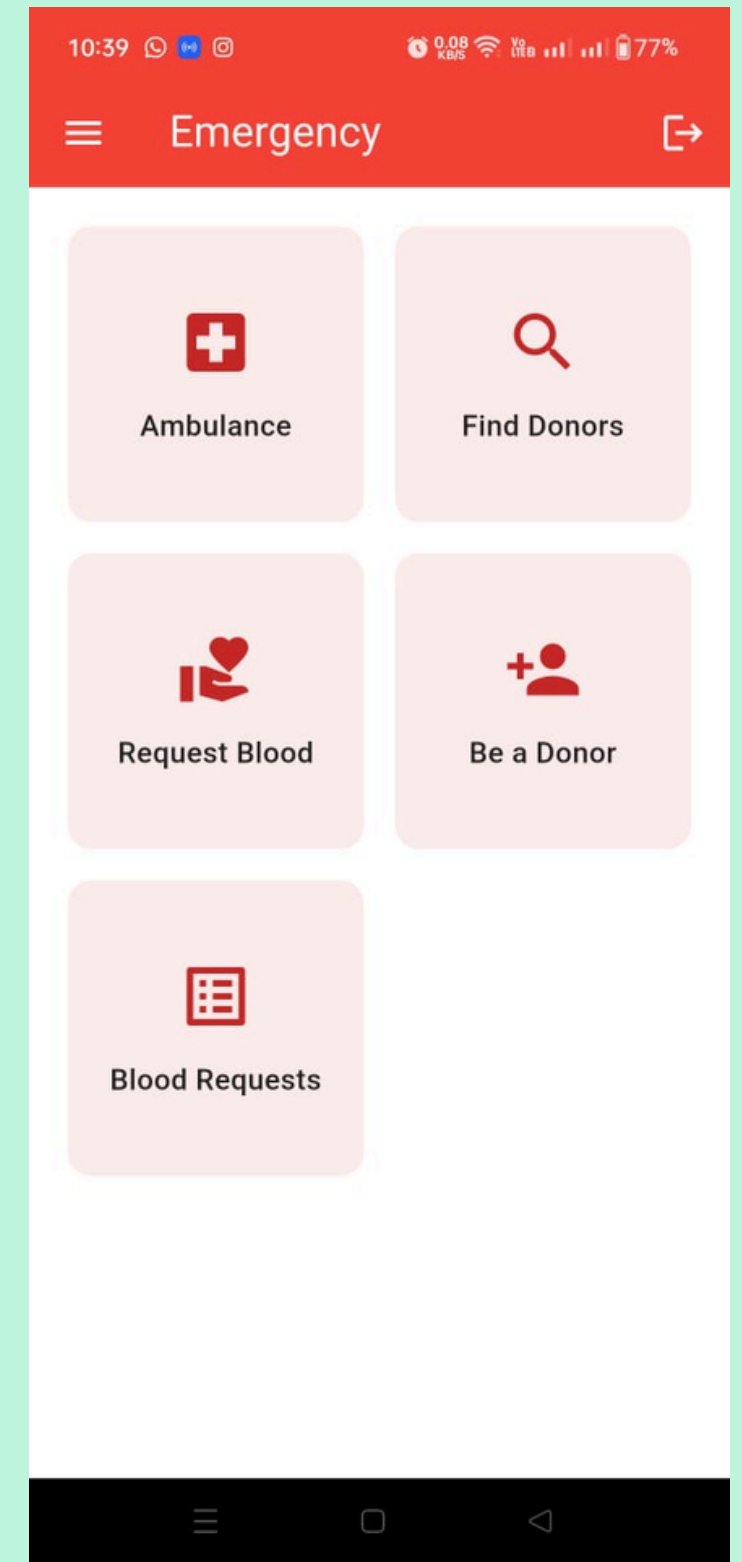
- Added an ambulance-request feature
- Implemented map integration
- Organized emergency features into a single section for better accessibility.



Campus map



Ambulance Page



Emergency Page



LIFELONG LEARNING:

Using the app, students and alumni can:

- Continuously interact even after graduation, ensuring lasting connections.
 - Apply app development knowledge in future real life projects.
 - Engage in lifelong learning through mentoring and knowledge-sharing opportunities.
- 



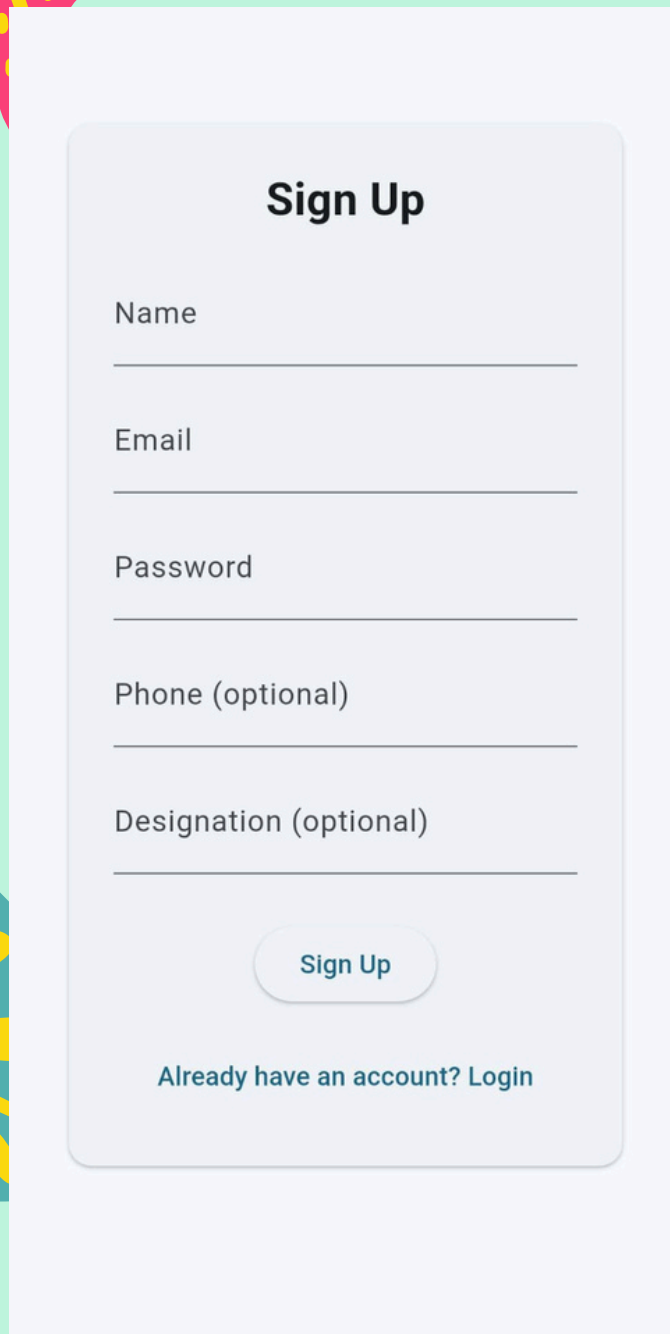
SUS SCORE

The System Usability Scale (SUS) evaluation yielded a score of 92.5, which falls in the “A / Excellent” range. The three benchmark tasks were: Map navigation
Emergency request, and Tuition/resources. The average task completion time was ~26 s based on two evaluators.



AUTHENTICATION & RESOURCE BANK

SADIA JAHAN MOON -202114085



A sign-up form with a light purple background. It contains fields for Name, Email, Password, Phone (optional), and Designation (optional). Below the fields is a 'Sign Up' button and a link 'Already have an account? Login'.

Sign Up

Name

Email

Password

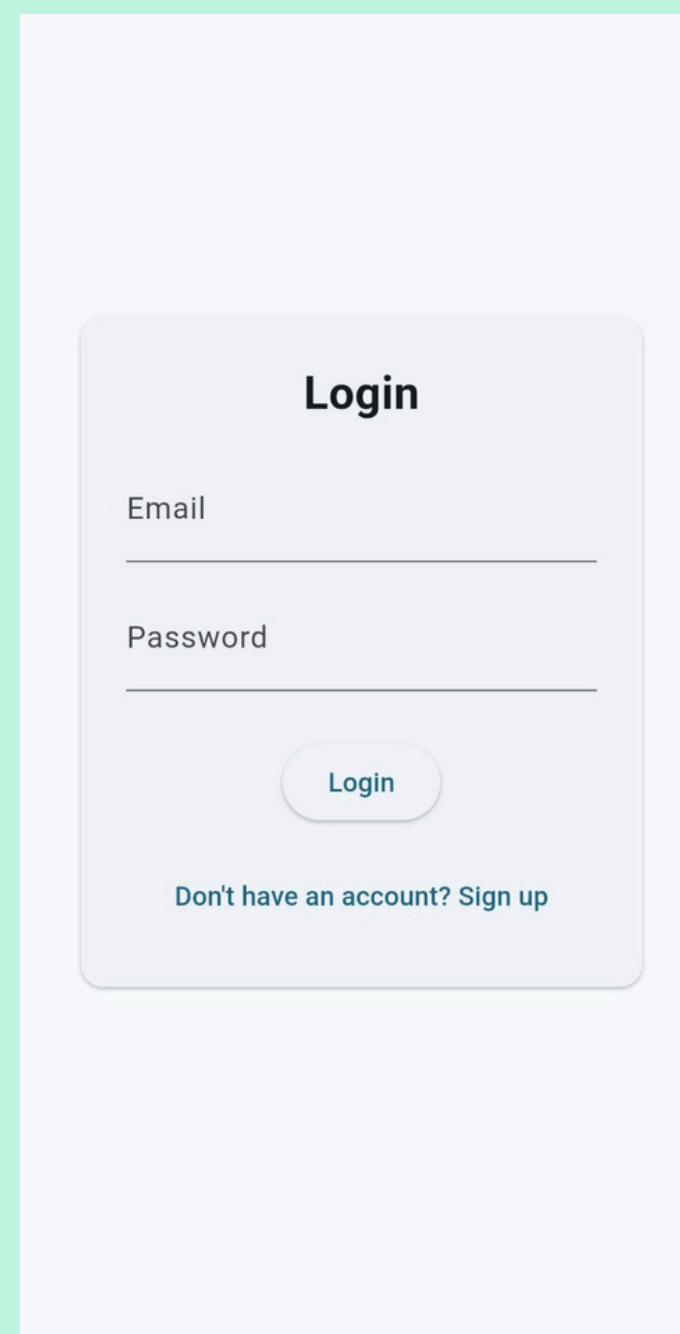
Phone (optional)

Designation (optional)

Sign Up

Already have an account? Login

Sign up page



A login form with a light purple background. It contains fields for Email and Password. Below the fields is a 'Login' button and a link 'Don't have an account? Sign up'.

Login

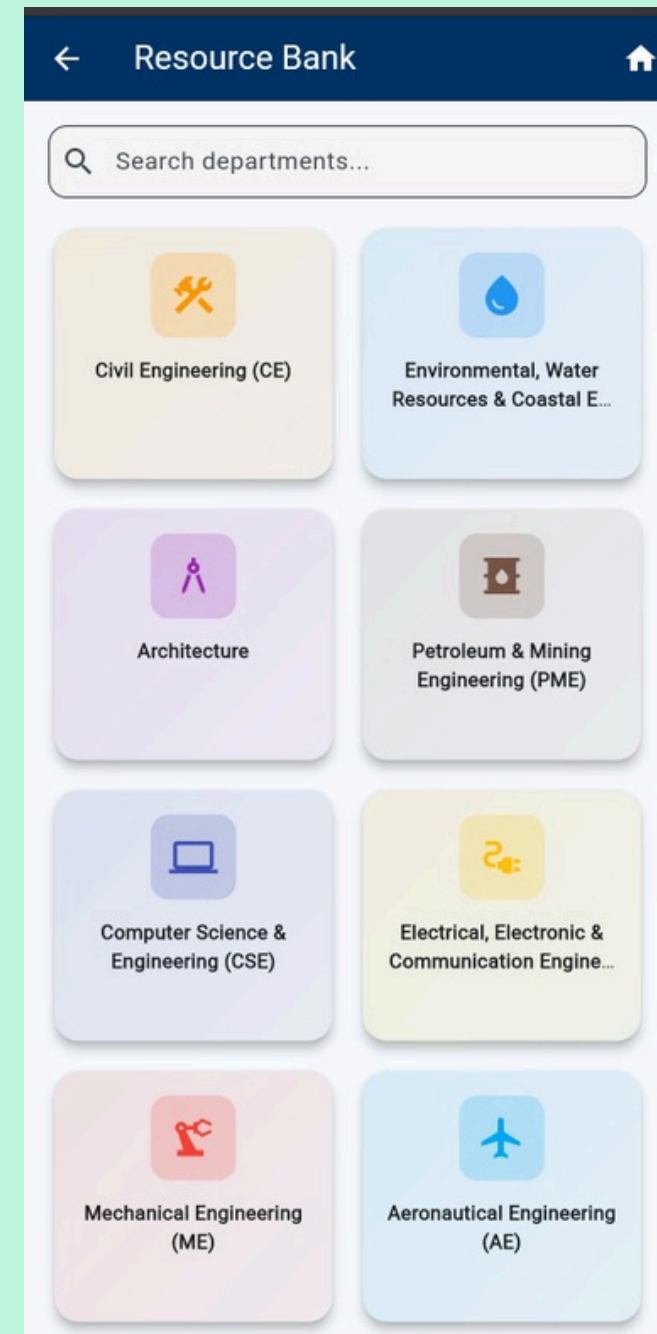
Email

Password

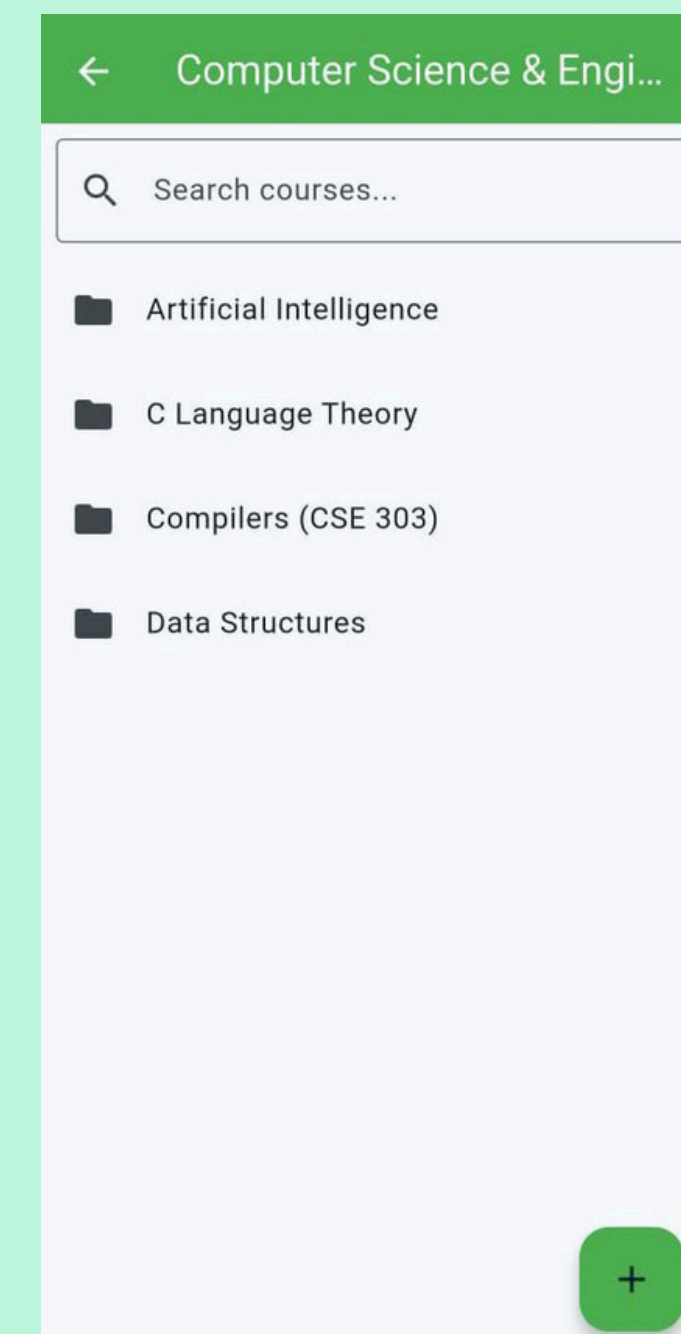
Login

Don't have an account? Sign up

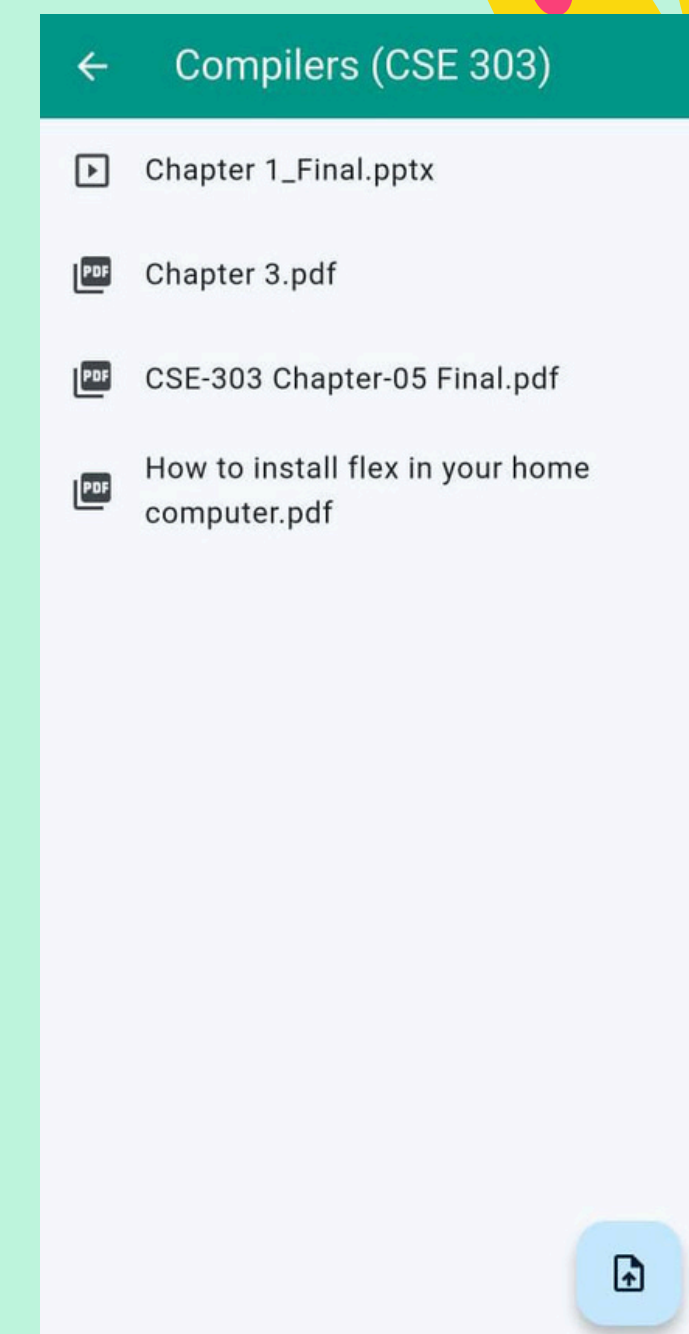
Login page



List of all departments



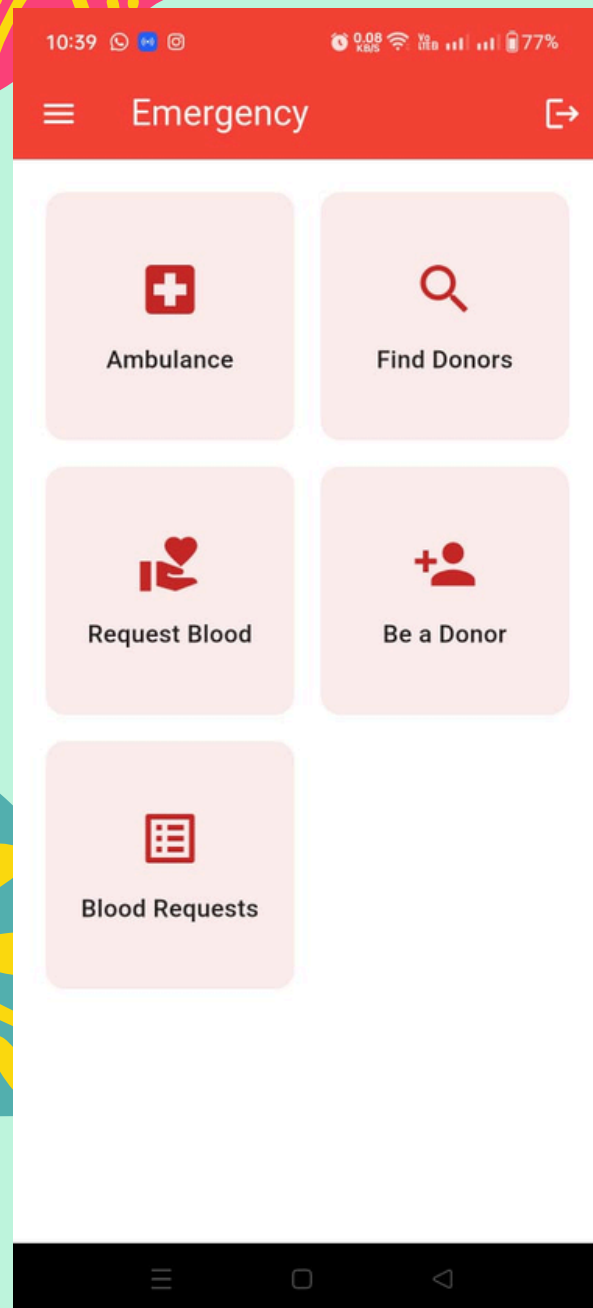
All courses of selected department



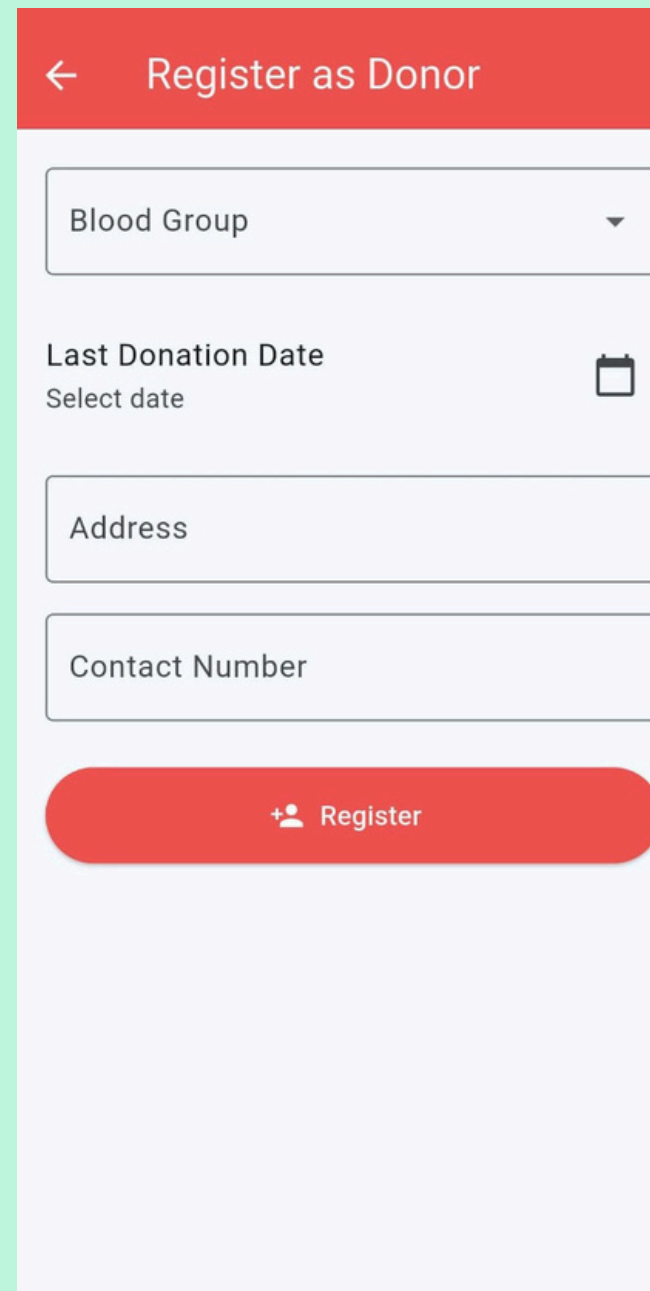
Chapters of that course

EMERGENCY

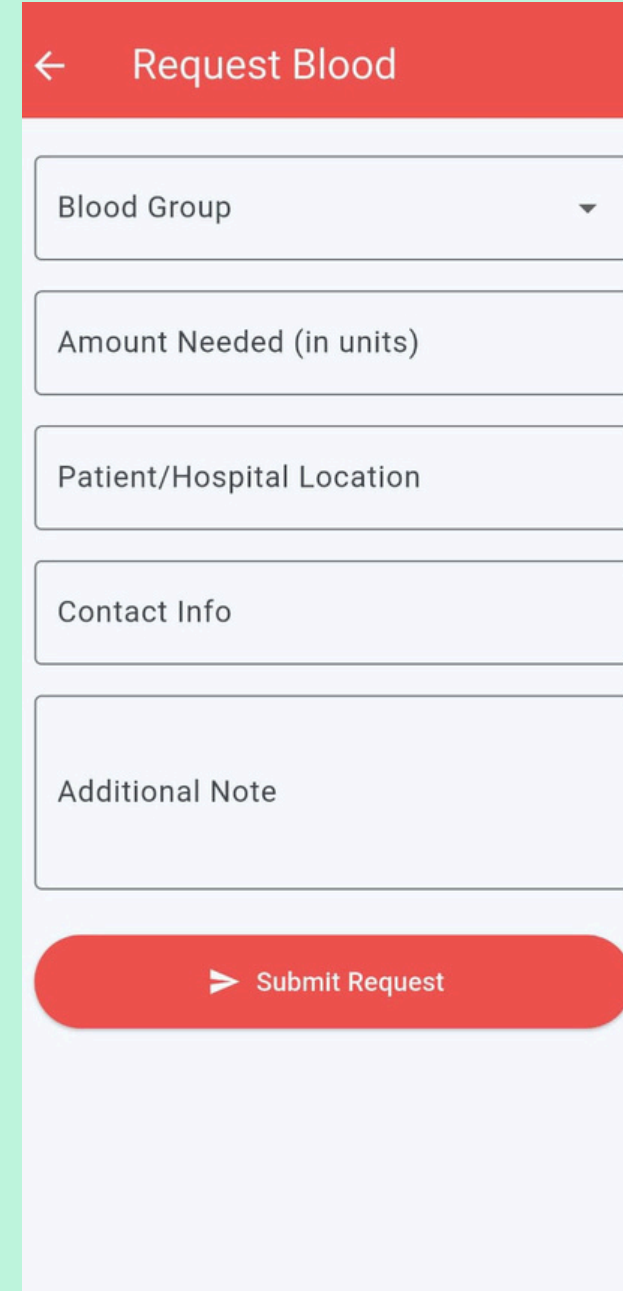
SADIA JAHAN MOON -202114085



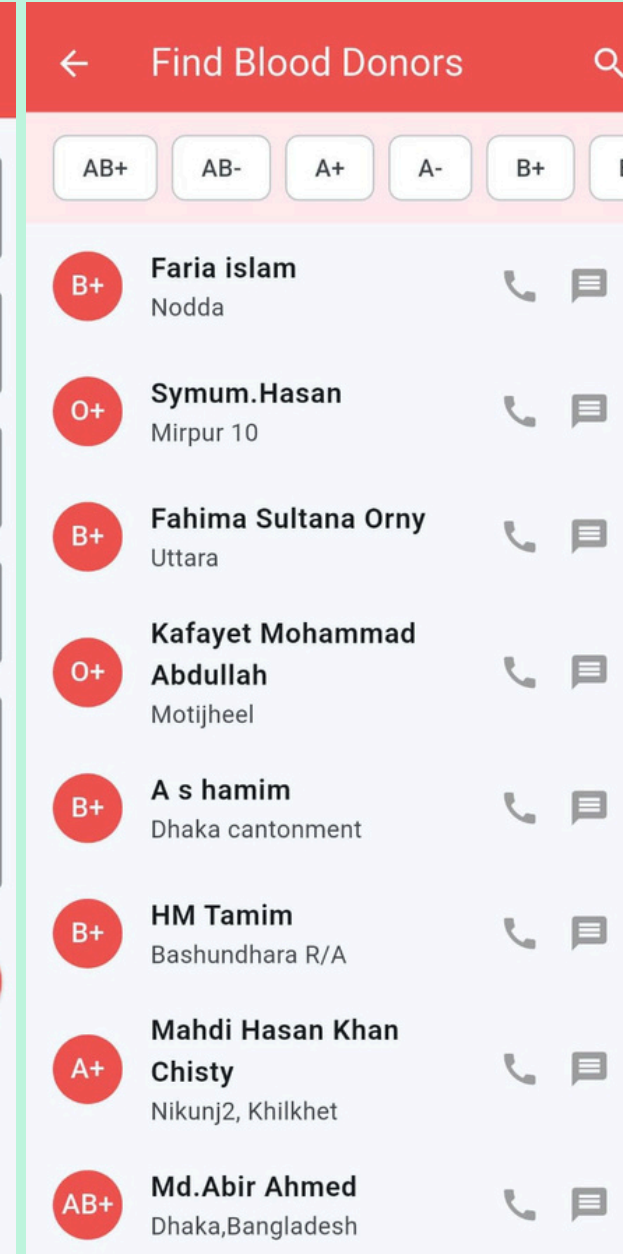
Emergency Home Page shows all the subfeatures



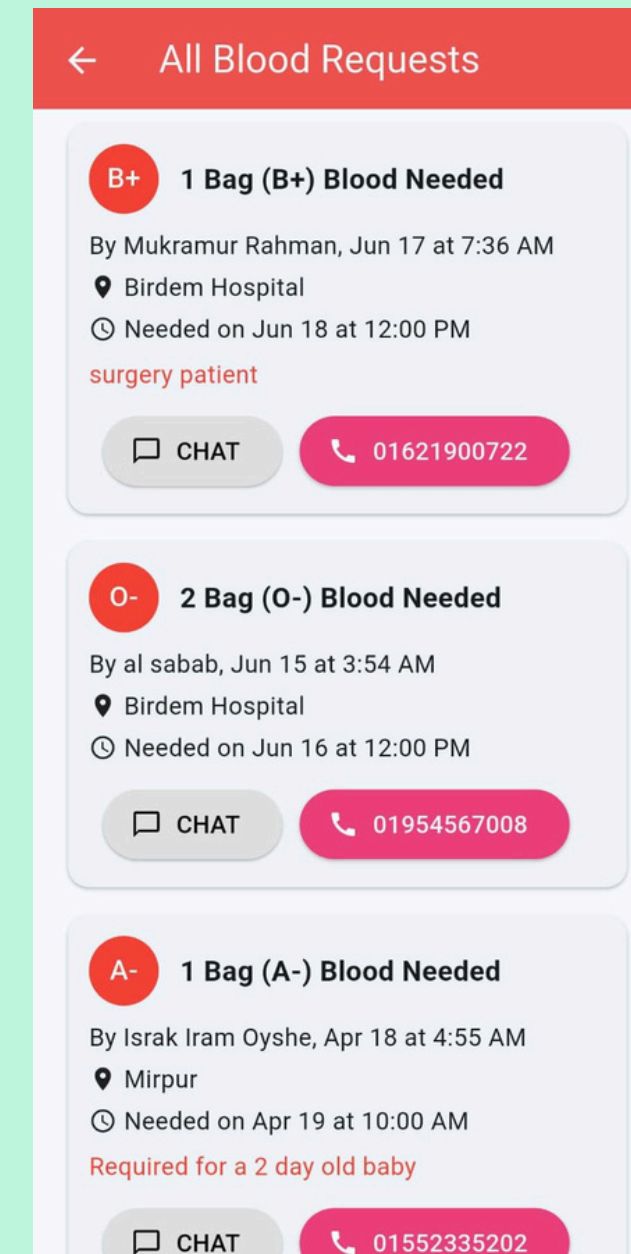
User can register as blood donor



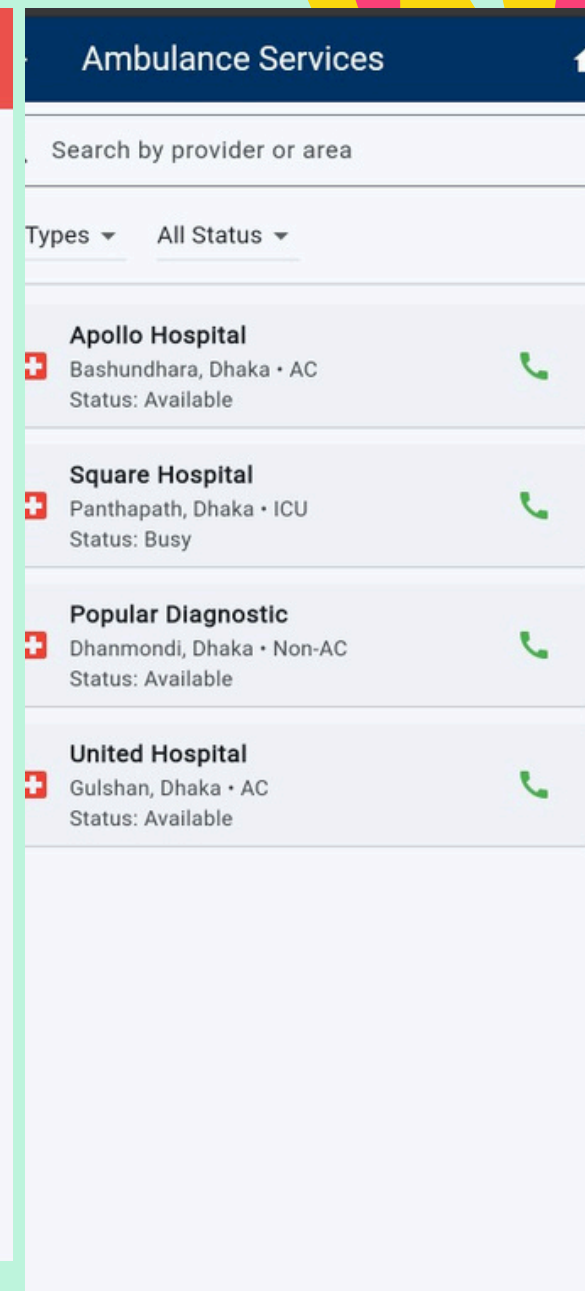
User can request for blood



List of all blood donors, user can search based on blood



List of all blood requests and details

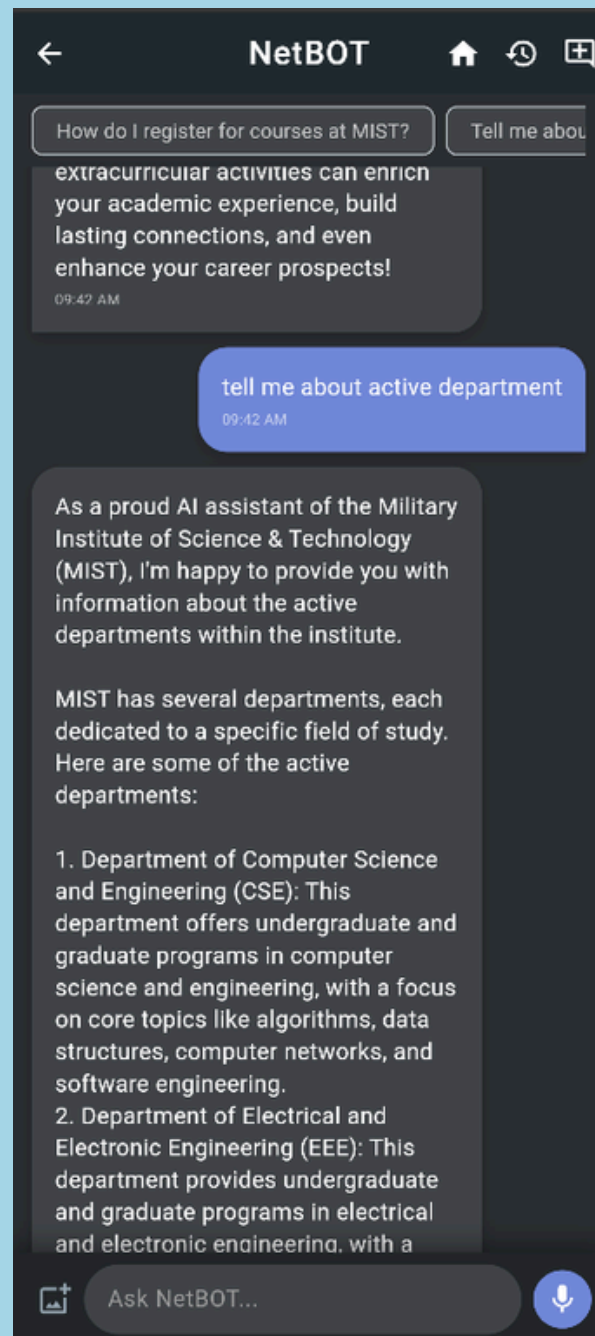


List of ambulance for emergency

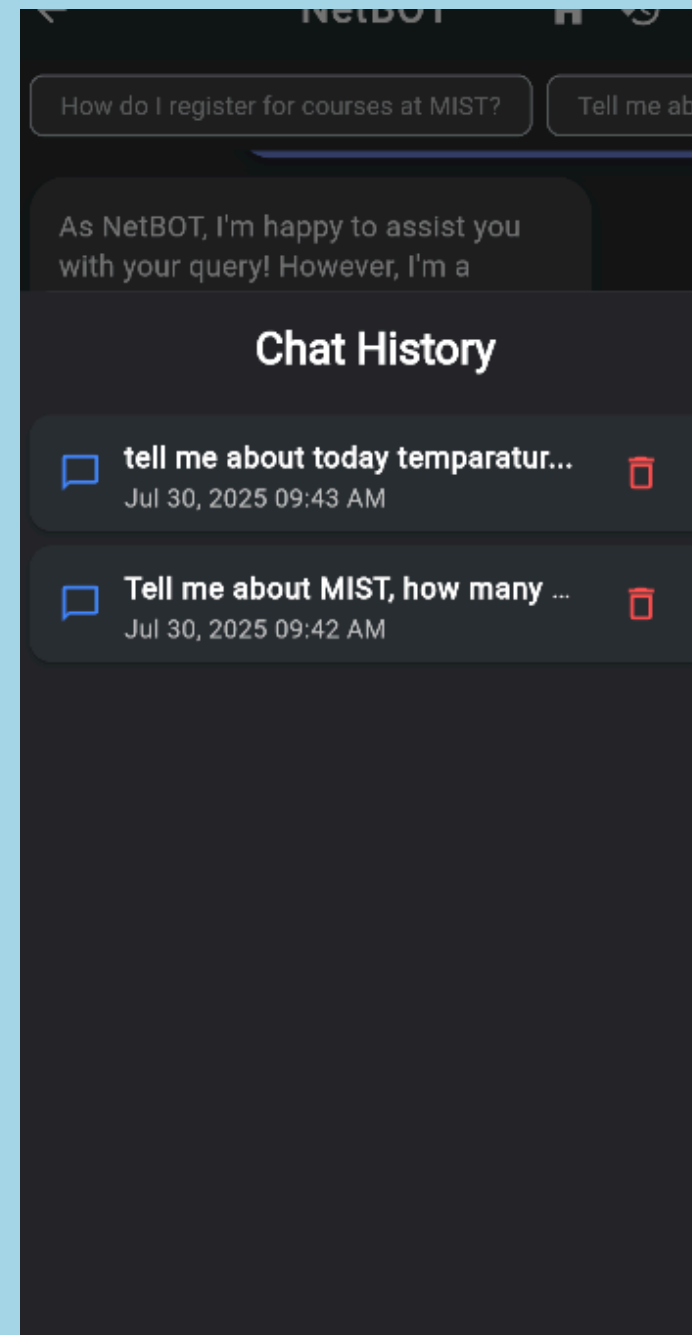
NETBOT

AUNINDYA PROSAD SAHA - 202114014

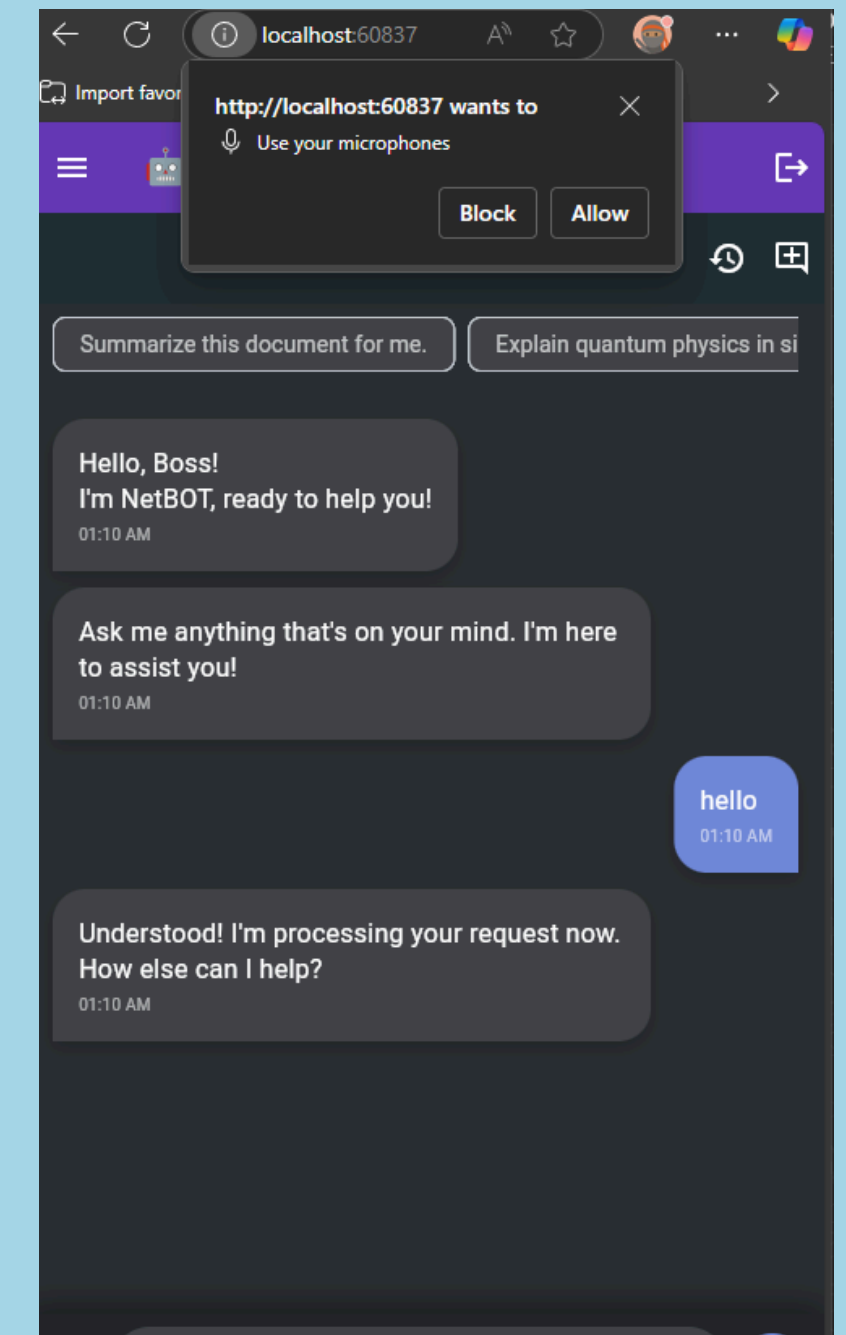
G. M. FAHIM TAZWAR - 202114025



ChatBot home page



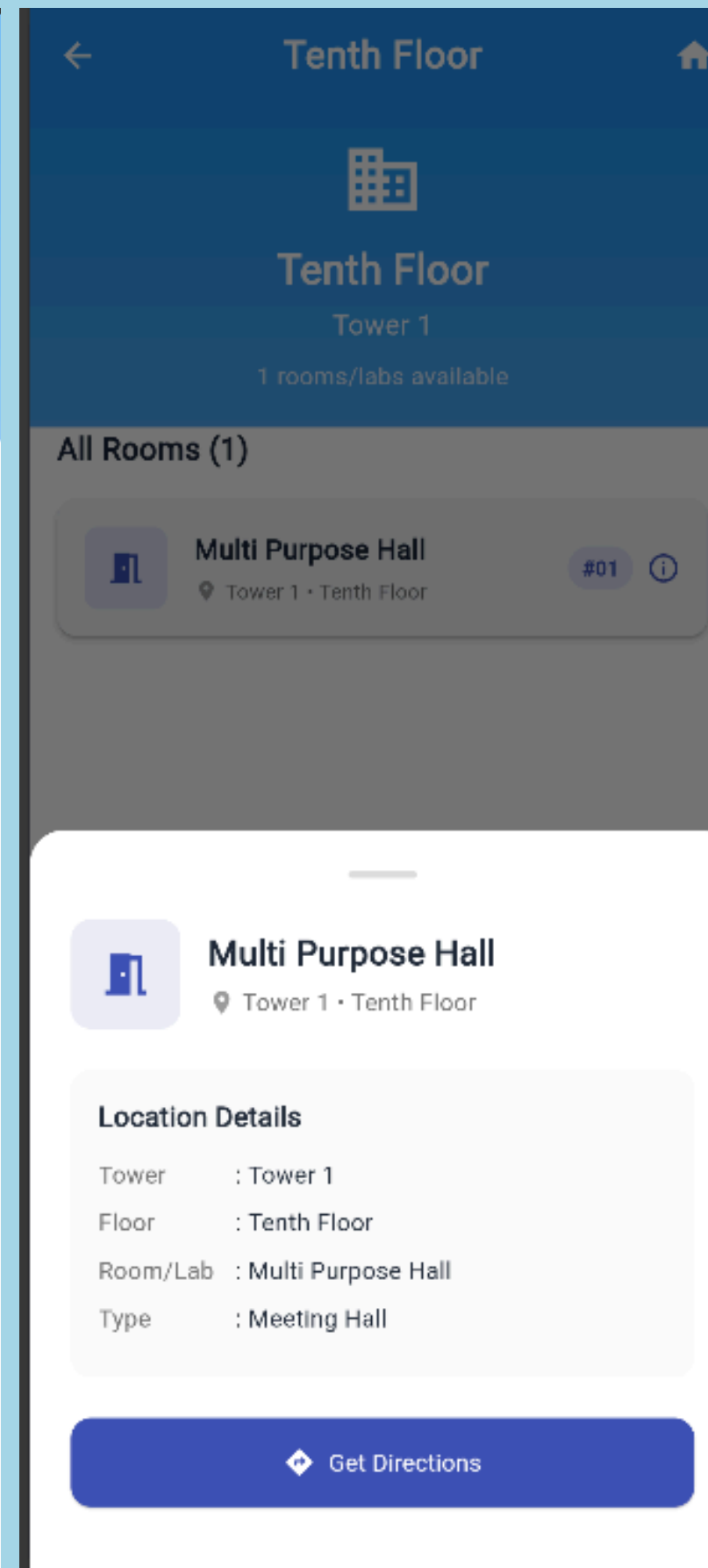
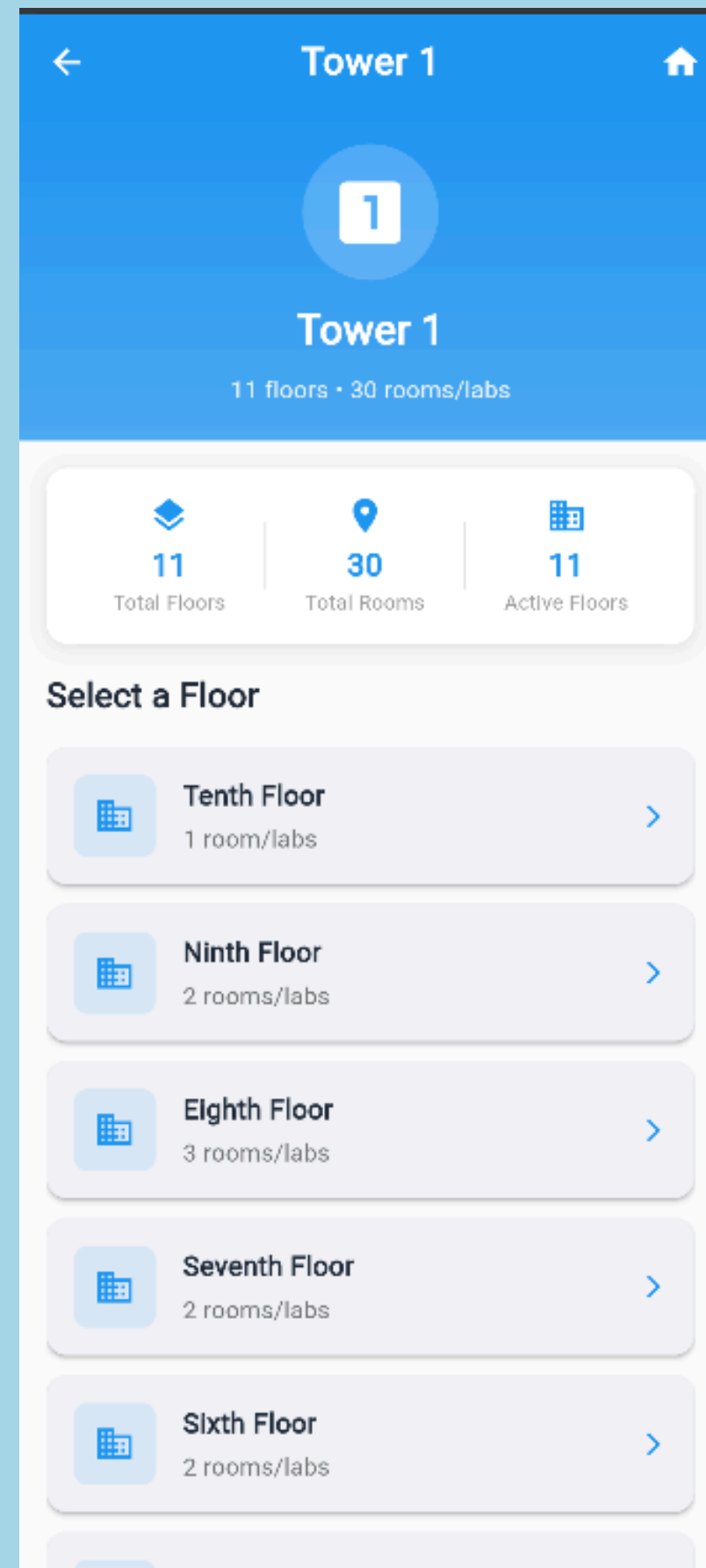
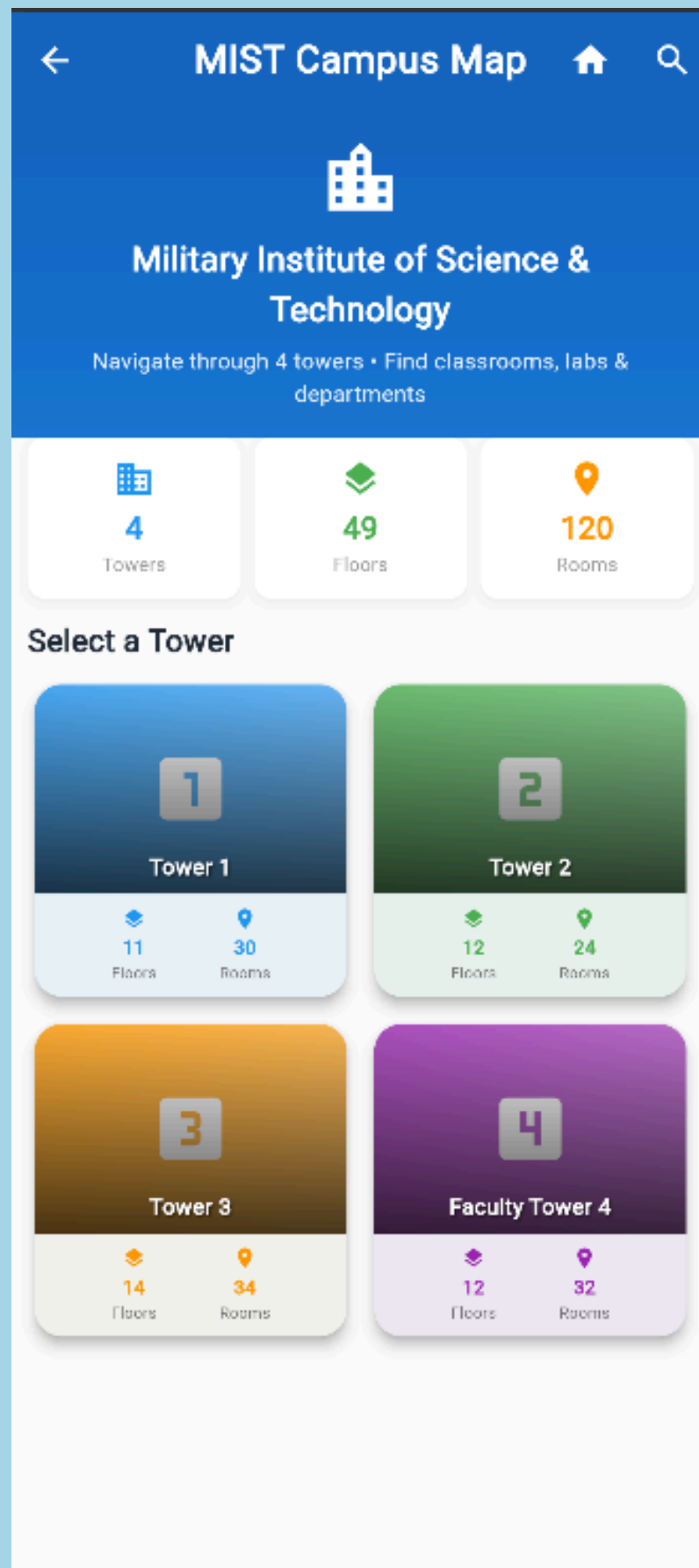
Chat Session History



Voice Input

CAMPUS MAP

AUNINDYA PROSAD SAHA - 202114014



Directions

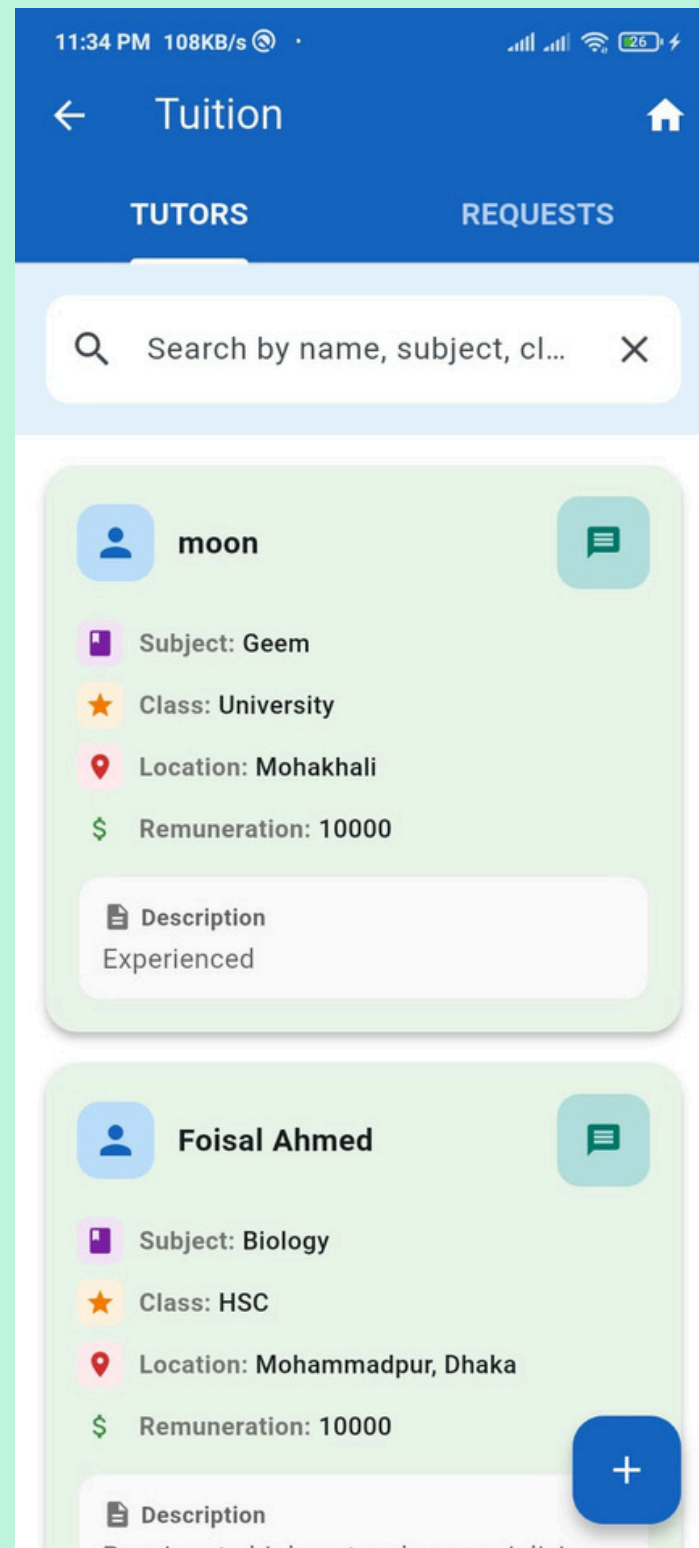
To reach: Multi Purpose Hall

1. Enter Tower 1
2. Take elevator/stairs to Tenth Floor
3. Look for room signage
4. Ask campus staff if needed

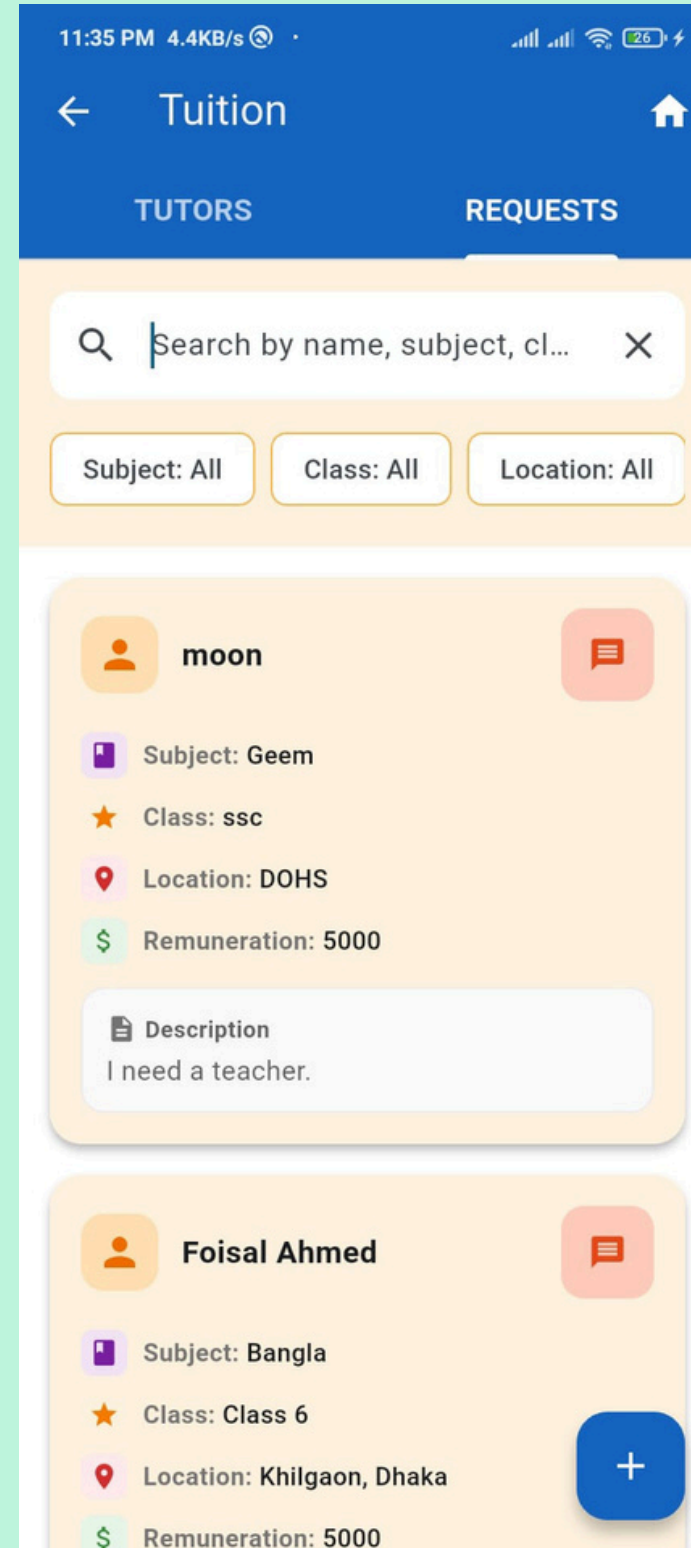
Got it

TUTION

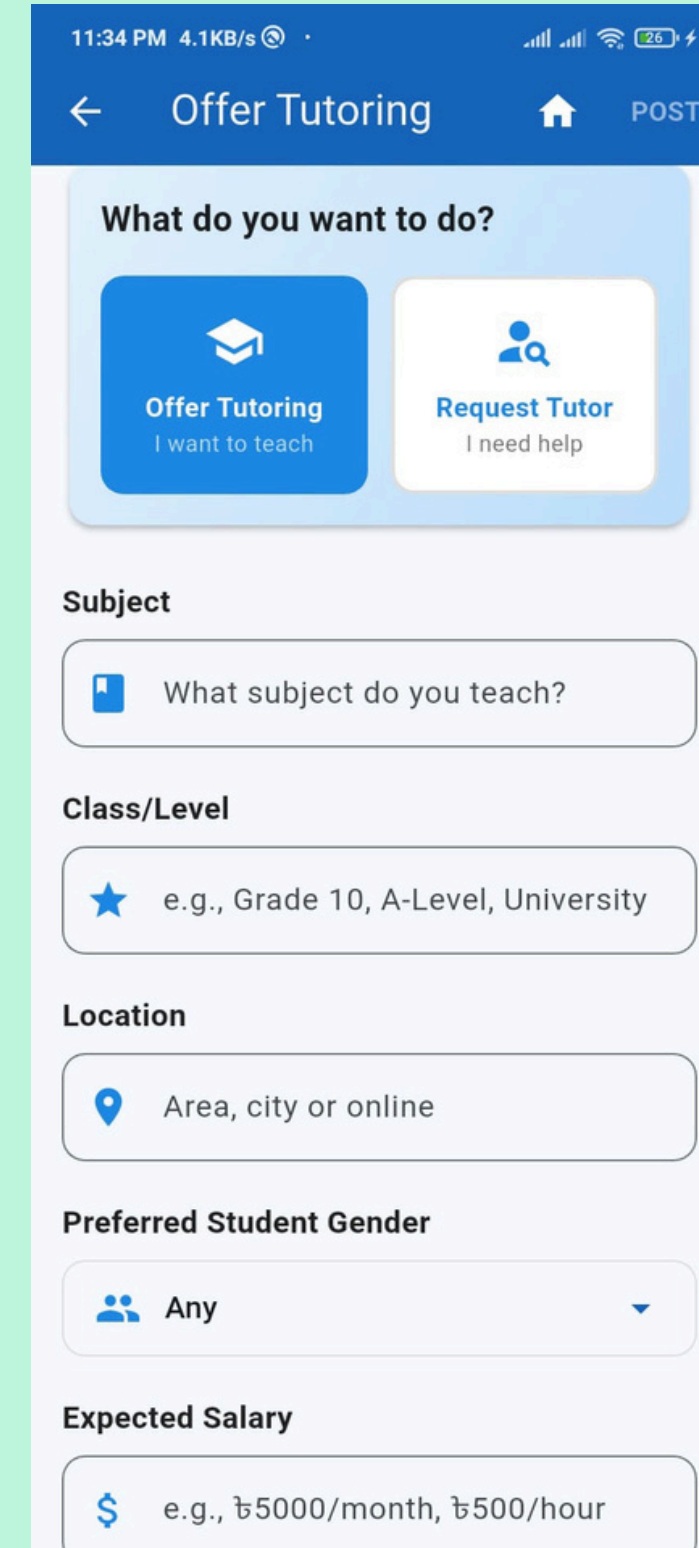
MEHEDI HASAN MOON - 202214048



Tutor Listings Page



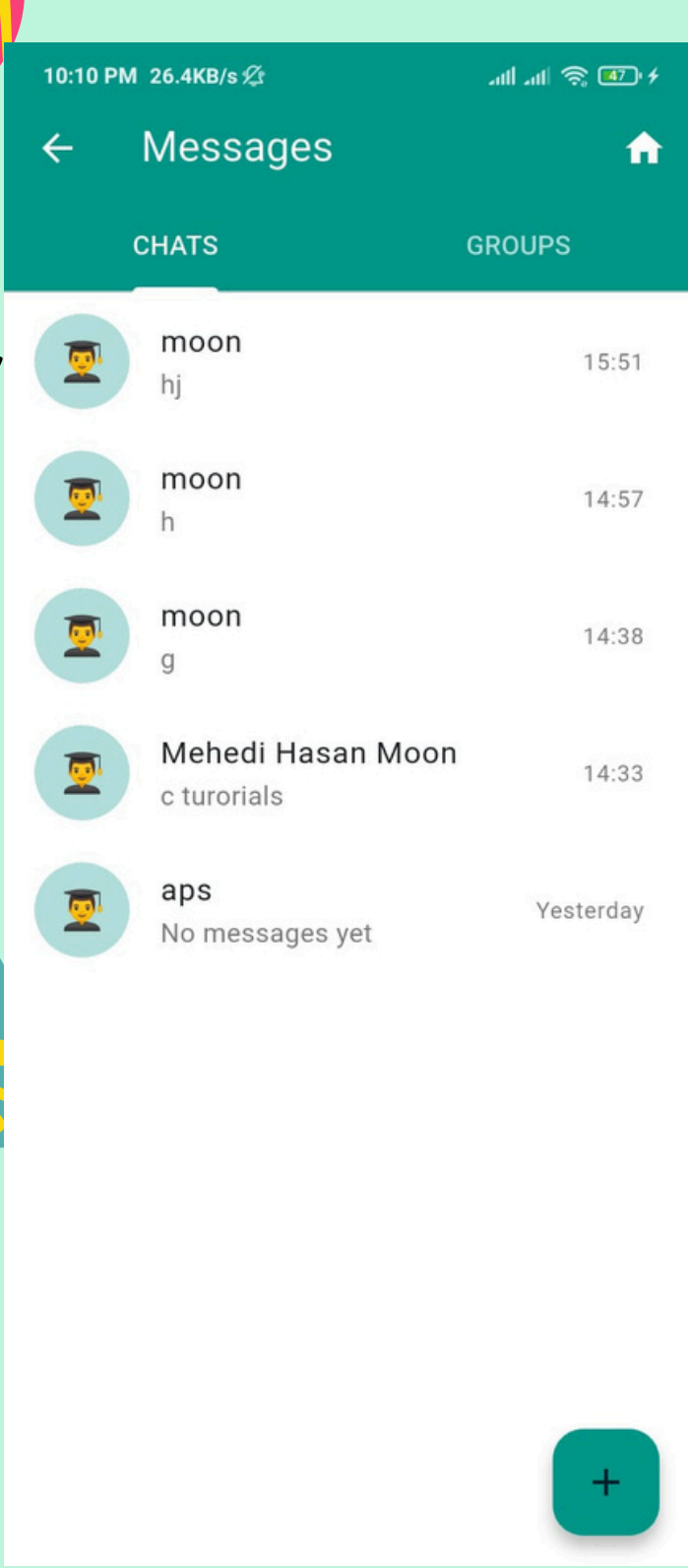
Tuition Requests



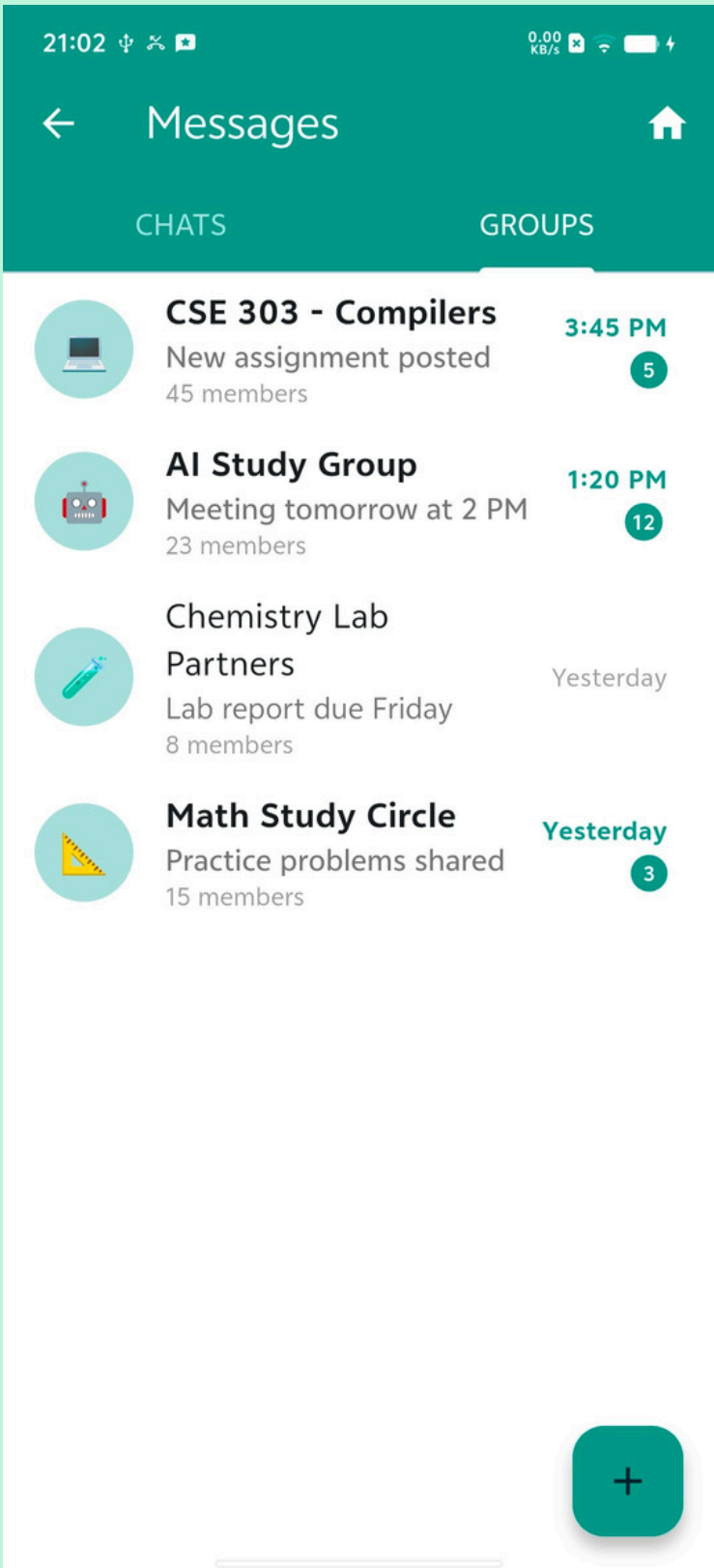
Post Tuition Page

MESSAGE

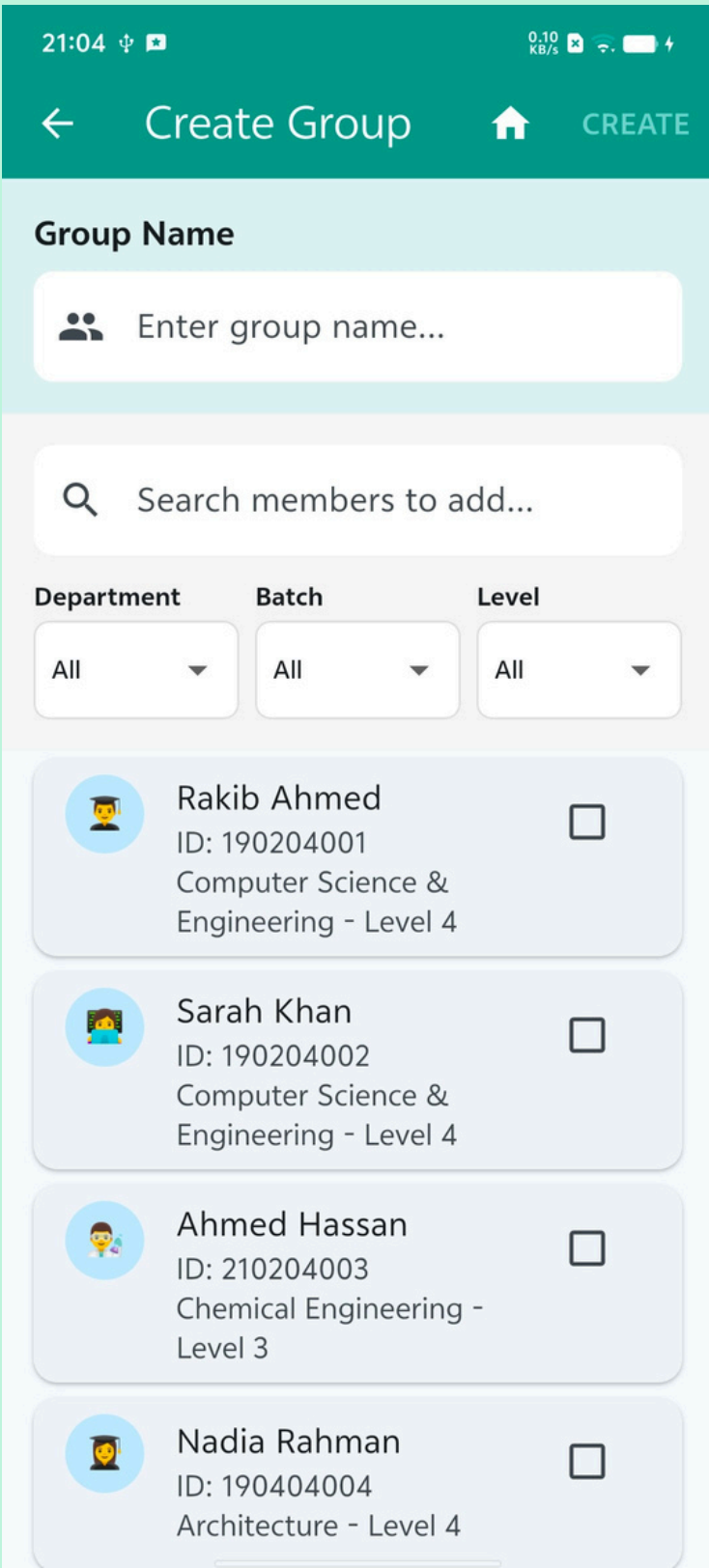
MEHEDI HASAN MOON - 202214048



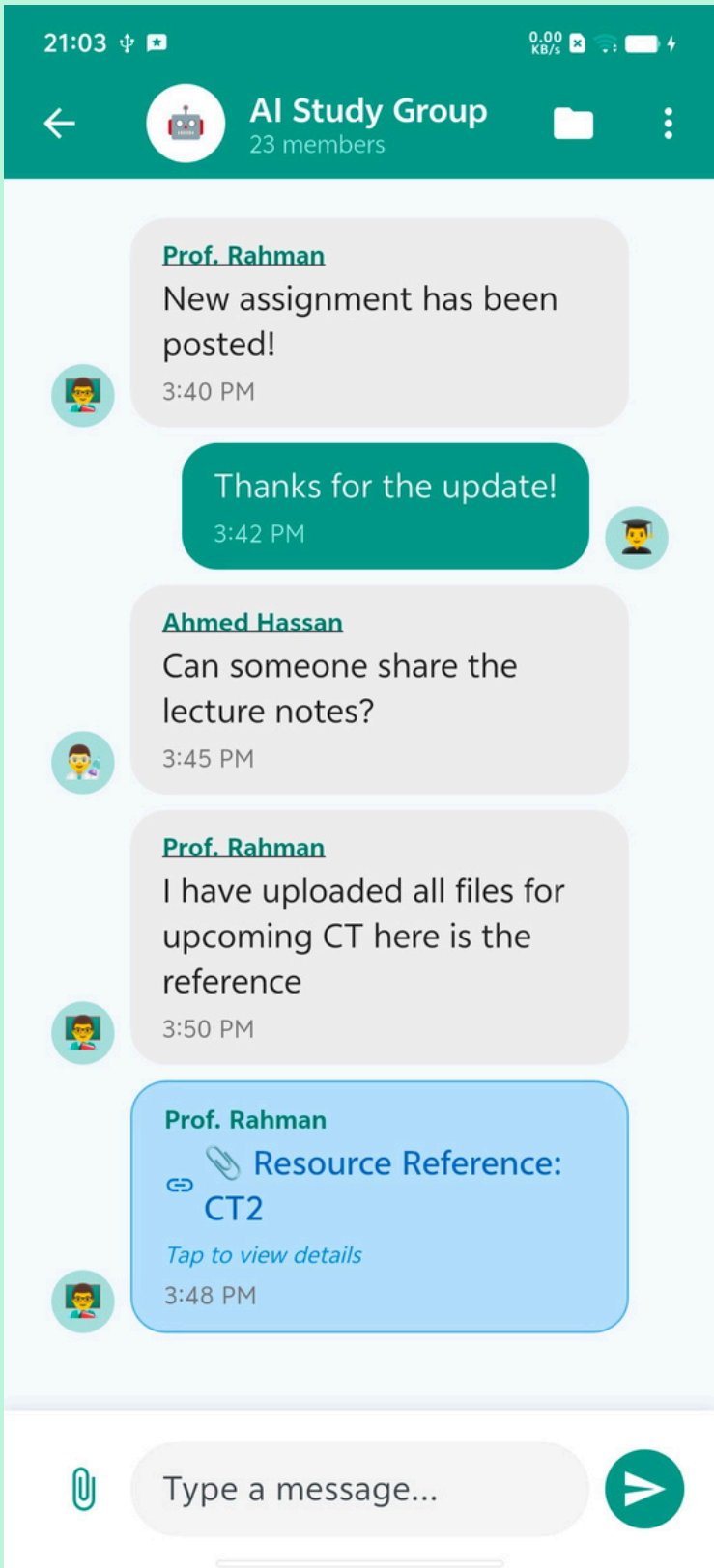
Chats



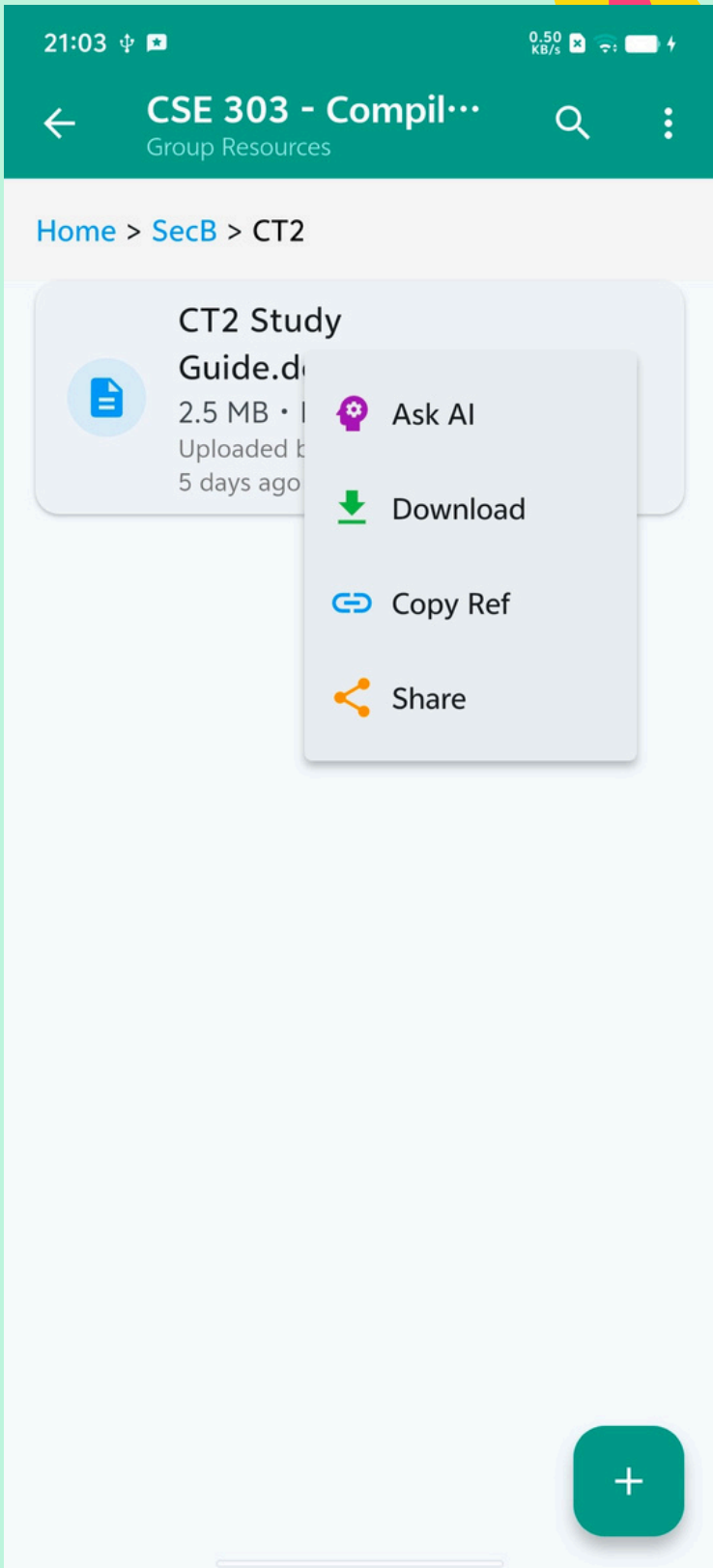
Groups



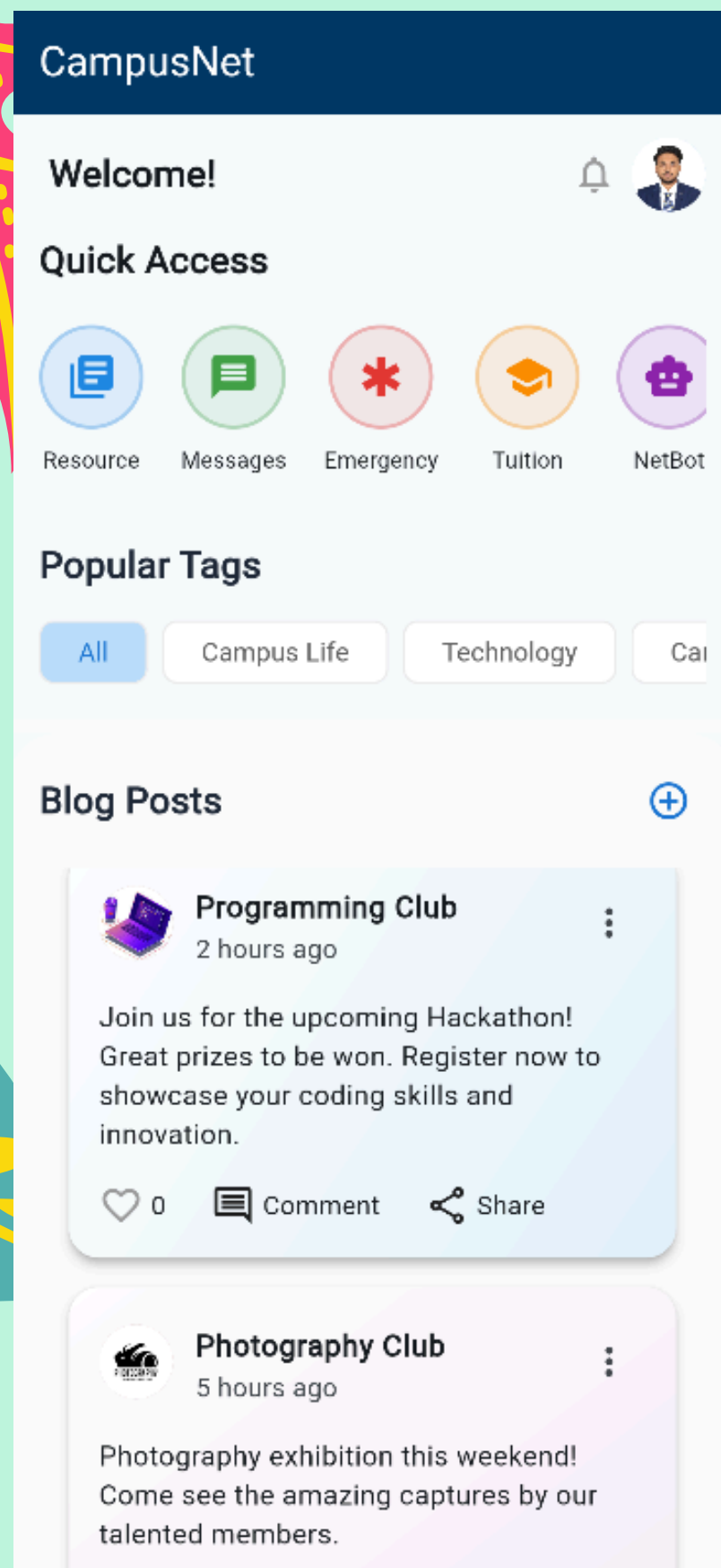
Create Chat/Group



Chats View



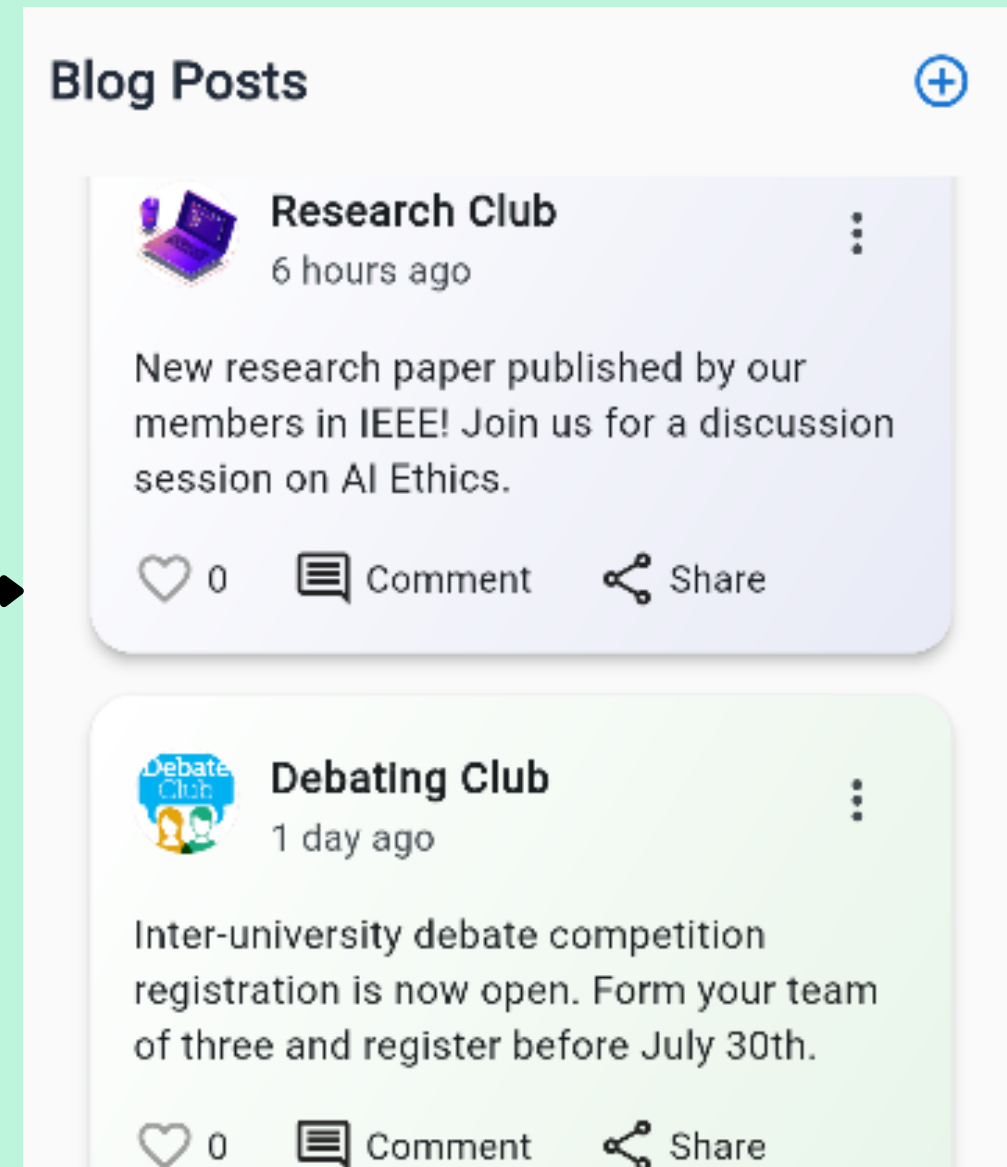
Group Resource



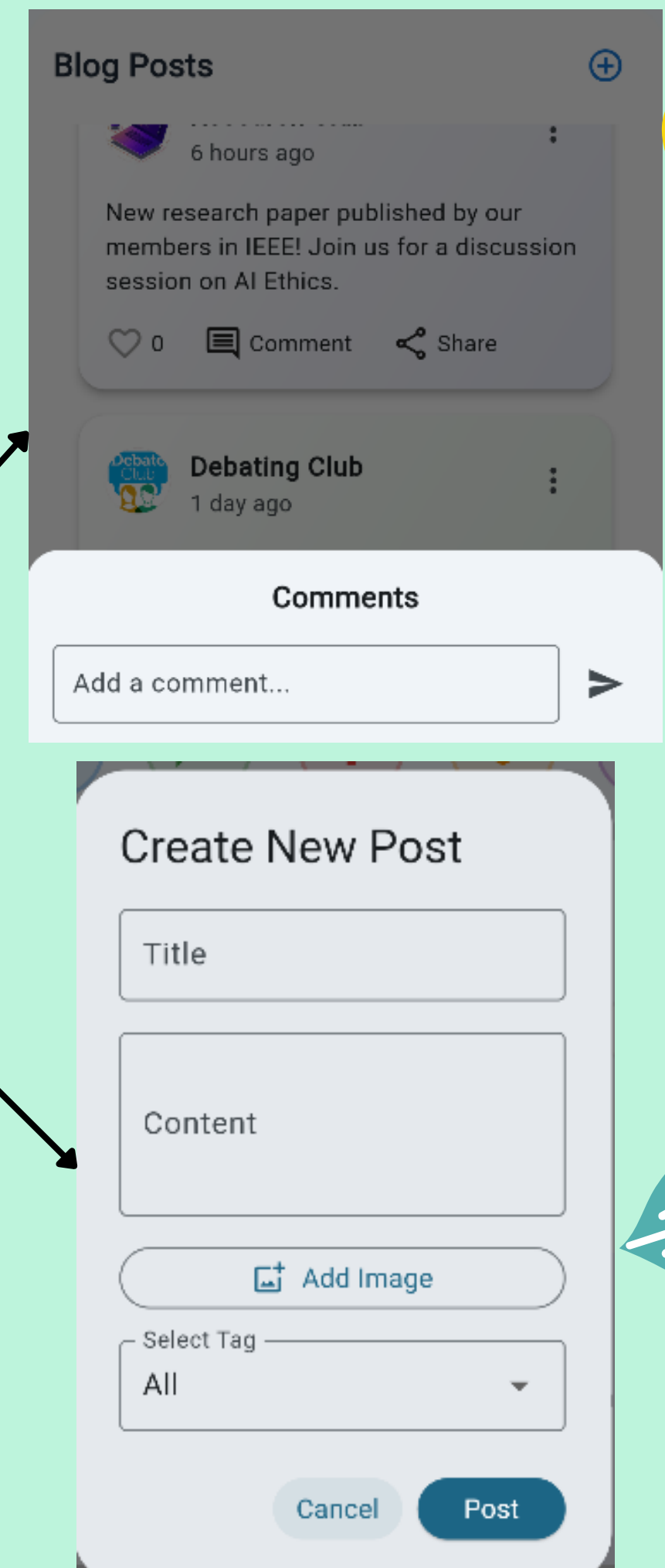
Acts as the landing screen where users can explore campus services

LANDING PAGE

G. M. FAHIM TAZWAR - 202114025



Displays all blog posts with titles, images, and summaries for users



PROFILE PAGE

G. M. FAHIM TAZWAR - 202114025

Displays
personal,
academic, and
contact details
of the user

 My Profile 



Fahim Tazwar Deesun
Undergraduate Student, CSE

 Email

fahim.deesun@mist.ac.bd

 Student ID

190204000X

 Department

Computer Science & Engineering

 University

Military Institute of Science & Technology

 Batch

2020

 Date of Birth

15 July 2002

 Phone

+8801XXXXXXXXXX

 Hometown

Dhaka, Bangladesh

 LinkedIn

[View Profile →](#)

 Facebook

[View Profile →](#)





Email

 fahim.deesun@mist.ac.bd

Phone

 +8801XXXXXXXXXX

Hometown / Address

 Dhaka, Bangladesh

Save Changes

REFERENCES

- [1] A. S. Khan, M. S. Nazir, and M. I. Khan, "Artificial Intelligence-based Chatbots for Education: A Review," International Journal of Educational Technology in Higher Education, vol. 17, no. 1, pp. 1-10, 2020. DOI: 10.1186/s41239-020-00214-0.
- [2] J. B. Smith and L. D. Miller, "Designing Mobile Applications for University Campuses: Challenges and Opportunities," Journal of Higher Education Research, vol. 25, pp. 45-58, 2019. DOI: 10.1016/j.jhedu.2018.12.008.
- [3] J. R. Hernandez, F. A. J. Ramos, and V. L. Garcia, "Developing Cross-Platform Mobile Applications with Flutter: A Practical Guide," Journal of Mobile Computing, vol. 7, no. 3, pp. 23-35, 2021. DOI: 10.1109/JMC.2021.010213.
- [4] M. H. Lee, J. Y. Kim, and S. H. Jang, "Resource Banks for Academic Success: A Case Study in University Learning Systems," Computers Education, vol. 128, pp. 245-259, 2019. DOI: 10.1016/j.compedu.2018.09.007.
- [5] T. A. Johnson, H. L. Zeng, and E. W. Nelson, "Enhancing Student Engagement Through Mobile Apps: A Study of University Students' Usage Patterns," Journal of Educational Technology, vol. 11, no. 4, pp. 89-102, 2020. DOI: 10.1016/j.educon.2020.06.003.
- [6] S. K. Lim, T. R. Lee, and R. D. Chang, "Developing Online Tutoring Systems for University Students: A Comprehensive Review," Journal of Internet Education, vol. 19, no. 2, pp. 74-86, 2021. DOI: 10.1016/j.jiedu.2020.10.005.
- [7] A. R. Kumar, S. N. Mehta, and R. L. Srivastava, "Campus Navigation Systems: A Review and Future Directions for Higher Education Institutions," International Journal of Computer Applications, vol. 139, no. 5, pp. 56-62, 2021. DOI: 10.5120/ijca202171672.



THANK YOU

Any Questions?

